Supplementary Material for "Reference History and Target Support in Multi-Window Photovoltaic Forecasting"

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1 Fixed-period extension and current verification scope

This addition uses saved expanded-grid coefficients without a new regularization search. Paired differences cover all five systems and full, power-active and low-power twelve-hour targets. Expanded B is compared with expanded A, Daily and each real Inverted seed; expanded A/B are also compared with their original-grid counterparts. The same draw is used across methods and seeds. Empty blocks remain in the sampling universe, and zero-support replicates and undefined skills are explicitly counted. NIST B is unchanged between grids; its zero difference is not discarded.

For Yulara twelve-hour/full, expanded B minus Daily is -0.224 kW with $[-1.814, 1.741]$ kW, while B minus mean Inverted is $+0.037$ kW with $[-1.175, 1.336]$ kW. These intervals cross zero without proving equivalence. The new-period analysis applies unchanged 2017 transformations and coefficients; two source buffers make every scheduled six-hour and previous-day input available when source observations exist. Each of the 26208 April–June origins is generated before applying target masks. The final target is 1 July 11:55. Available source/operation metadata do not establish that no equipment change occurred. Missingness and transformed input ranges are reported independently of model score.

1.1 Alice restored support: unresolved numerical replay

The frozen neural Train routine accepts complete nonnegative H144 labels without requiring a valid origin label; Ridge does require a valid origin. For Qcells this gives neural Train/Validation counts 12747/2999 and Ridge 12648/2977. They are different supports, not interchangeable counts. The provider materials reviewed do not define all finite negative power as invalid. Keeping the historical rule provides traceability, not proof of its physical justification.

All 36 Alice checkpoints were loaded and evaluated without training. Sixteen reproduced the historical predictions within the original 2×10^{-5} relative and absolute tolerances; twenty did not. The original neural preprocessing objects were not separately saved. A saved later Train-only processor reconstructs the current historical window builder exactly, but that is not a direct comparison with saved historical neural inputs. Backend and batch diagnostics did not resolve every discrepancy. Restored-origin prediction arrays are retained as unaccepted diagnostics; no formal all-model S1/S2 score or ranking is asserted. The support counts below are valid independently of that pending numerical issue.

S1 scores individual finite nonnegative targets, and S2 scores finite raw power, while retaining the historical nonnegative Last-value and Daily definitions. At Qcells twelve hours, historical Daily-matched support has 2996 origins and 431150 pairs; S1 has 6786 origins and 958668 pairs, and S2 has 6786 origins and 959382 pairs. Active counts are 187950, 442887 and 442887, respectively. These are overlapping forecast pairs, not independent observations. Their unique target timestamps are 6425, 6781 and 6786. This enlarged scoring population cannot establish the performance of a differently trained model.

1.2 Current versus historical checks

Six external Inverted checkpoints reproduced the complete original 2017 prediction arrays before the 2018 period was evaluated. Original and expanded external Ridge predictions also reproduced. Fit methods and weight serialization were prohibited at runtime. New-period metric and additive-MSE reductions, block replays, protected-file inventories and PDF checks are recorded with their actual scope. Earlier M1-R/M2 suite totals remain historical records, not newly repeated suites. No neural model was trained in this extension.

2 Diagnostic revision: evidence and interpretation

This supplement places the later diagnostic methods and explanatory figures before the complete historical tables. The neural experiments and the original-grid Ridge evidence are preserved. New regularization fits are post hoc sensitivities on already observed performance splits, not new prespecified confirmation. File-based tests below concern numerical and bookkeeping consistency, not a count of independent experiments.

2.1 Yulara actual-matrix diagnosis

The saved Train-fitted processor, design means/scales, coefficients, target transformation, and timestamps reconstruct the original Test arrays. The objective is $\|Y - XB\|_F^2 + \alpha\|B\|_F^2$ after standardization and target centering, with an unpenalized intercept. A Cholesky solve uses the same actual 69333-by-504 or 69333-by-792 matrix. For the original selected parameters 0.1 and 1000, the maximum Test difference from the saved spectral solution is below 9×10^{-9} kW. Normal-equation relative residuals and objective comparisons are supplied, together with the full per-feature Train scales and later offsets.

The origin-position weather-missing flags occur in four eligible Train windows. Their small scale, 0.007595, maps an observed missing flag to about 132 standardized units. Validation exposure is 18.56% and Test exposure 5.50%. The two weather masks coincide and their prediction components are correlated; their separate RMS components must not be added as independent MSE contributions. These linear components and the concentration of errors in a small group of windows support a missing-pattern extrapolation explanation. No window is removed and no prediction is clipped. Diagnostic quantiles use every finite target in saved candidate trajectories; headline twelve-hour metrics use the stricter complete-H144 and Daily-finite intersection.

Seven original alpha choices lie at grid endpoints. After diagnosing the actual matrices, a common $10^{-4}, \ldots, 10^{8}$ grid was fixed for all five systems and A/B variants. The same original Train transformations and full-H144 Validation objective select each parameter. Every Validation candidate is retained. The expanded choices and Test errors are supplementary sensitivity results, with the original grid still reported. Detailed solver records distinguish original-model verification from low-alpha numerical checks in the supplemental fits.

2.2 Qcells exclusion attribution

Every five-minute coordinate within a split is a candidate origin. Overlapping flags identify insufficient input history or target tail, future missing/nonfinite labels, finite negative future labels, invalid origin power and unavailable Daily points. Exclusive accounting gives boundary precedence, then future nonfinite, future negative, origin invalid and retention. Daily is a separate point intersection; it is not used to explain primary origin deletion. Tables distinguish origin counts, origin–lead target pairs and unique physical target timestamps.

For Test one-hour support, complete-twelve-hour conditioning removes 3467 origins: 117 boundary cases and 3350 future-negative cases under the exclusive order. The negative flag overlaps 65 of those boundary cases, giving 3415 flagged origins overall. There are no future nonfinite-label exclusions. Fifty-four raw negative records (52 at hour 18, two at hour 7) affect many overlapping long trajectories. Their values range from approximately -0.010933 kW to numerical values near zero; no physical or invalid-code interpretation is inferred. The frozen rule remains unchanged. Train and Validation exhibit the same selection process. Original neural Train/Validation label-completeness counts can differ from evaluation because the historical Alice fitting routine does not require a valid origin power.

2.3 Ridge paired temporal effects

All comparisons use complete H144 origins and an identical Daily-finite target intersection across A, B, Daily and Inverted seeds 42/43/44. Full, power-active and low-power use the same unchanged Train threshold. Blocks are anchored at the Test split's midnight in the frozen coordinate and assigned by forecast origin, not target time. Fixed EST is retained at NIST; Yulara has provider-local coordinates. Empty blocks remain in the resampling universe. The 48-hour main scheme and 24/72-hour sensitivity use 2000 draws and random seed 20260908.

Each draw sums block SSE and valid counts, then computes RMSE. B-minus-reference is negative when B is better. Each real neural seed is compared separately; the mean contrast averages per-seed effects under the same draw. Deterministic Ridge has no artificial seed repetitions. The 48-hour Test series contains 12 Alice blocks or 46 external blocks; some power subsets have fewer nonempty blocks. Full block counts, SSE, actual target-index support and effects are in the lightweight evidence. Overlapping trajectories can still couple adjacent blocks, and intervals condition on fitted predictions and observed dates. They do not include retraining or guarantee unobserved-year performance.

2.4 Actual versus historical validation

This diagnostic revision checks saved Ridge predictions against reconstructed matrices. It compares solver objectives and residuals, exact timestamp alignment, Qcells exclusion accounting and paired block arithmetic. SHA-256 inventories compare the original data, frozen results, checkpoints and original prediction arrays before and after. The old M1-R/M2 full suites and original neural checkpoint reproduction remain historical records, not new claims. Memory chunking and datetime-resolution normalization in the diagnostic code do not alter the fixed EST or provider-local coordinate. No neural training occurs.

At the expanded NIST A choice of 10^{-4}, an independent augmented QR solve has full column rank 504. Its maximum prediction difference from the spectral solution is 0.002390 kW; the full-scope RMSE difference is -1.835×10^{-7} kW. This small-alpha numerical sensitivity is retained in the diagnostic evidence, rather than replacing either prediction array. It does not affect the original-grid NIST B contrasts reported in the main text.

3 Post hoc review: support, uncertainty and linear references

These analyses were planned after the original performance results were known; they do not change the frozen neural experiments. H12/H48/H96/H144 are cumulative 1/4/8/12 h windows. Common-H144 comparisons retain only complete twelve-hour origins and evaluate every prefix there. Fixed-lead curves score individual leads on that same support. Daily comparisons use the same elementwise target intersection for every method. The tables in the accompanying review CSVs include both origin and target counts for every support.

Power-active targets have true power above 1% of Train maximum; low-power is the complement, not night. In that preceding review, all 2400 tested full/active/low-power squared-error identities passed on saved neural and reference arrays. Full MSE is the sum of subset SSE divided by the full target count. The per-seed decomposition file also records MAE, bias, counts and each weighted MSE contribution. Its neural summaries must average per-seed metrics rather than square an averaged RMSE.

The paired block procedure uses fixed 24/48/72 h nonoverlapping origin blocks anchored at Test midnight, 2000 resamples, and random state 20260908. Forty-eight hours is the primary block length. Empty blocks are retained. Methods and seeds share each draw. Per-seed RMSE is recomputed from summed SSE and count; the reported statistic is the mean of per-seed reference skill. Percentile intervals condition on the fitted models and observed dates. Dependence can persist across block boundaries, and Alice has much shorter temporal support than the external sites. No p-values or independent-comparison interpretation are attached to win counts.

Ridge A uses 72 times the neural input width in flattened, Train-standardized features (1224 at Alice and 504 externally). Ridge B adds 144 lagged power values at target time minus 24 h and 144 missingness indicators, giving 1512 or 792 features. Train medians fill missing lag inputs. The same Train-only numerical imputer, masks, Isolation Forest and neural feature transformations precede these linear designs. A separate standardizer is fitted to each final design. The target transformation is fitted to valid Train power. A single regularization parameter per system and variant is selected from 0.1, 1, 10, 100, 1000 on complete H144 Validation global MSE. Deterministic spectral ridge solutions include an unpenalized intercept; no Test values choose regularization. All ten fitted references and all 50 Validation candidate scores are retained, including the unfavorable Yulara results.

Monthly summaries assign complete H144 origins to October, November or December. Fixed cases use the first eligible origin at or after the 15th at 10:00. Additional best/worst cases explicitly use post hoc seed42 Daily-error differences on complete Daily-valid trajectories. Each displayed twelve-hour curve belongs to one origin. Neither weather conditions nor operation events are inferred from appearance.

The preceding September 8 review re-read all 60 training histories and verified each recorded best Validation epoch. It independently checked 480 Ridge metric rows, all ten selected parameters and finite prediction shapes, and five synthetic spectral/direct solver equivalences: 565 checks passed, zero failed or skipped. The 43/17/10 historical protocol/artifact suites and 10432 M2 evidence comparisons reported above are historical audit results, not newly repeated full suites in this review. No neural fit, checkpoint update or prediction regeneration was performed. A further 6720 metric/count comparisons between the new point-array reductions and the unchanged historical CSVs passed, with zero failures or skips. Eight additional synthetic forward checks confirmed both input widths and participation of parameter-owning modules, without loading checkpoints. The saved-prediction audits and their explicit scope accompany the review materials.

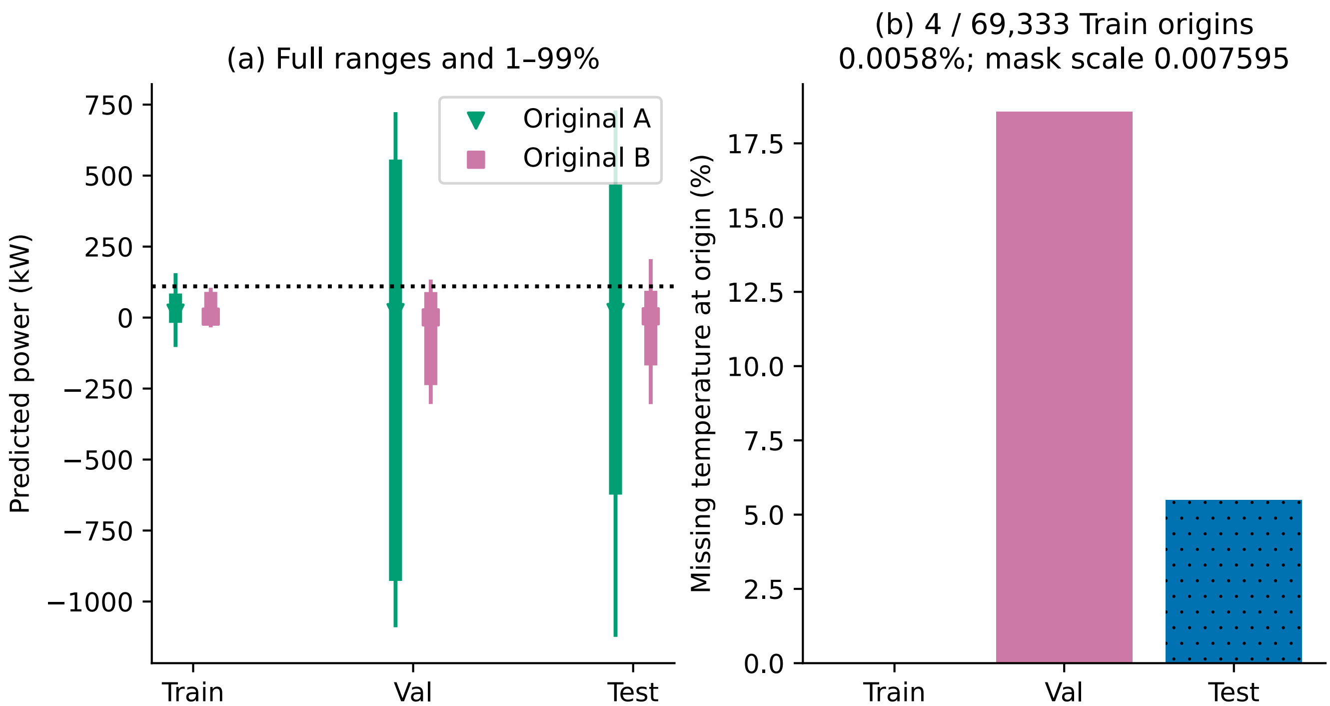

Figure S1: Yulara original Ridge distributions and weather-missing exposure: 4/69333 Train origins (0.0058%). Thin lines show extrema, thick lines 1st–99th percentiles, markers medians. The dotted line is the Train target maximum, not a clipping threshold.

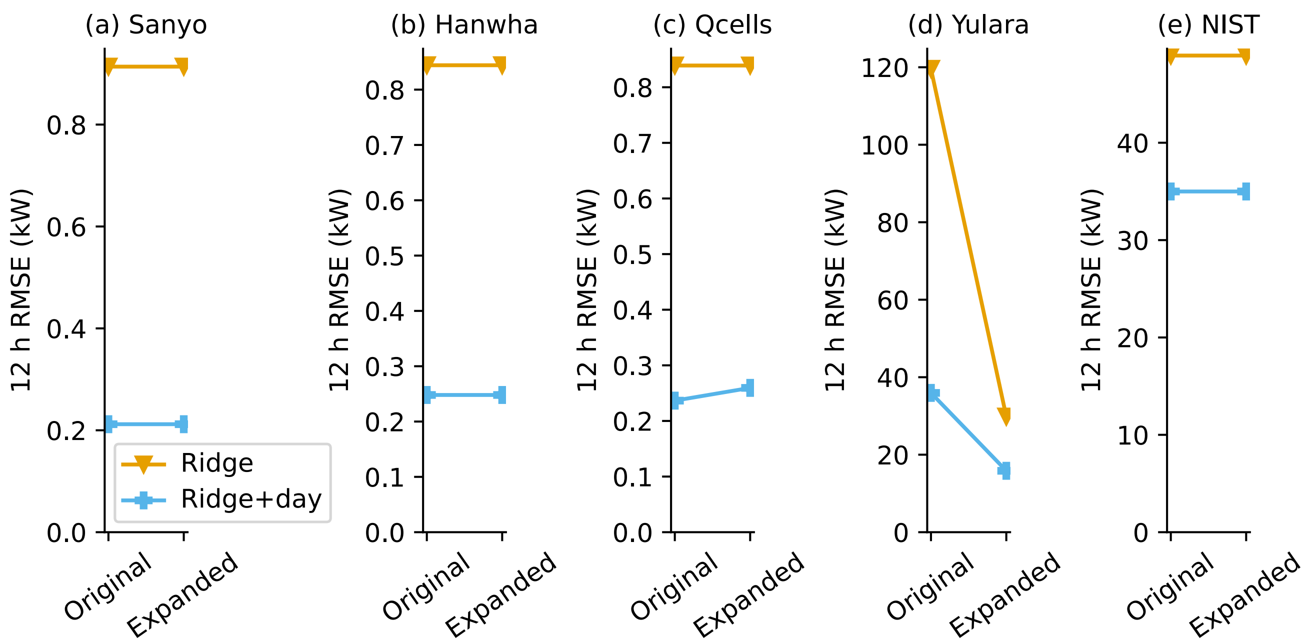

Figure S2: Uniform expanded alpha grid 10^{-4} through 10^{8}, selected solely by full-H144 Validation MSE for all five systems and both information sets. Original and expanded Test errors use identical Daily-matched full targets; all ten comparisons are shown.

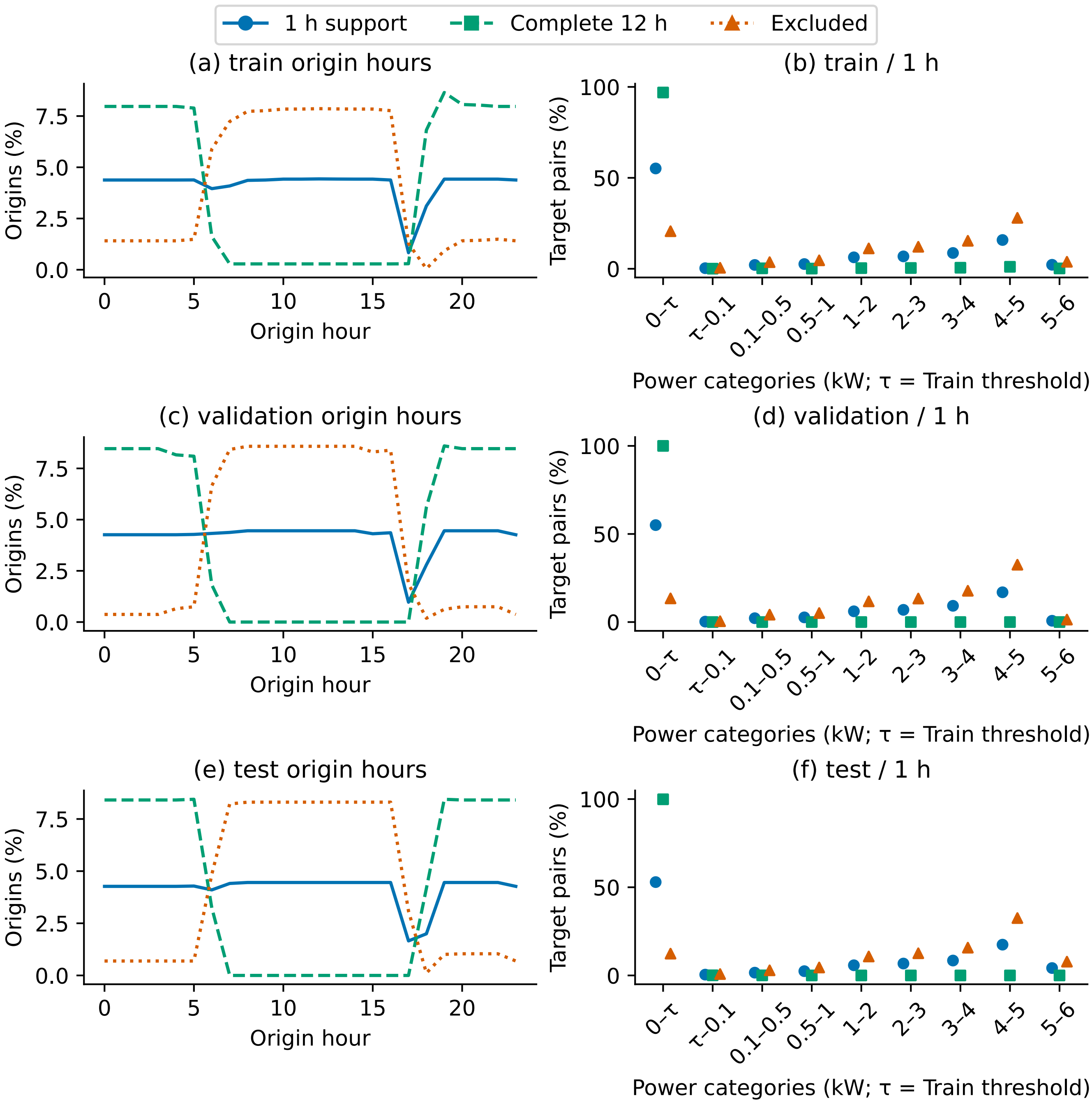

Figure S3: Qcells support composition in every split. Left: origin-hour proportions. Right: categorical points for unequal-width power bins, normalized by pairs in the displayed bins. Complete-H144 conditioning selects different hours and power ranges in Train and Validation as well as Test.

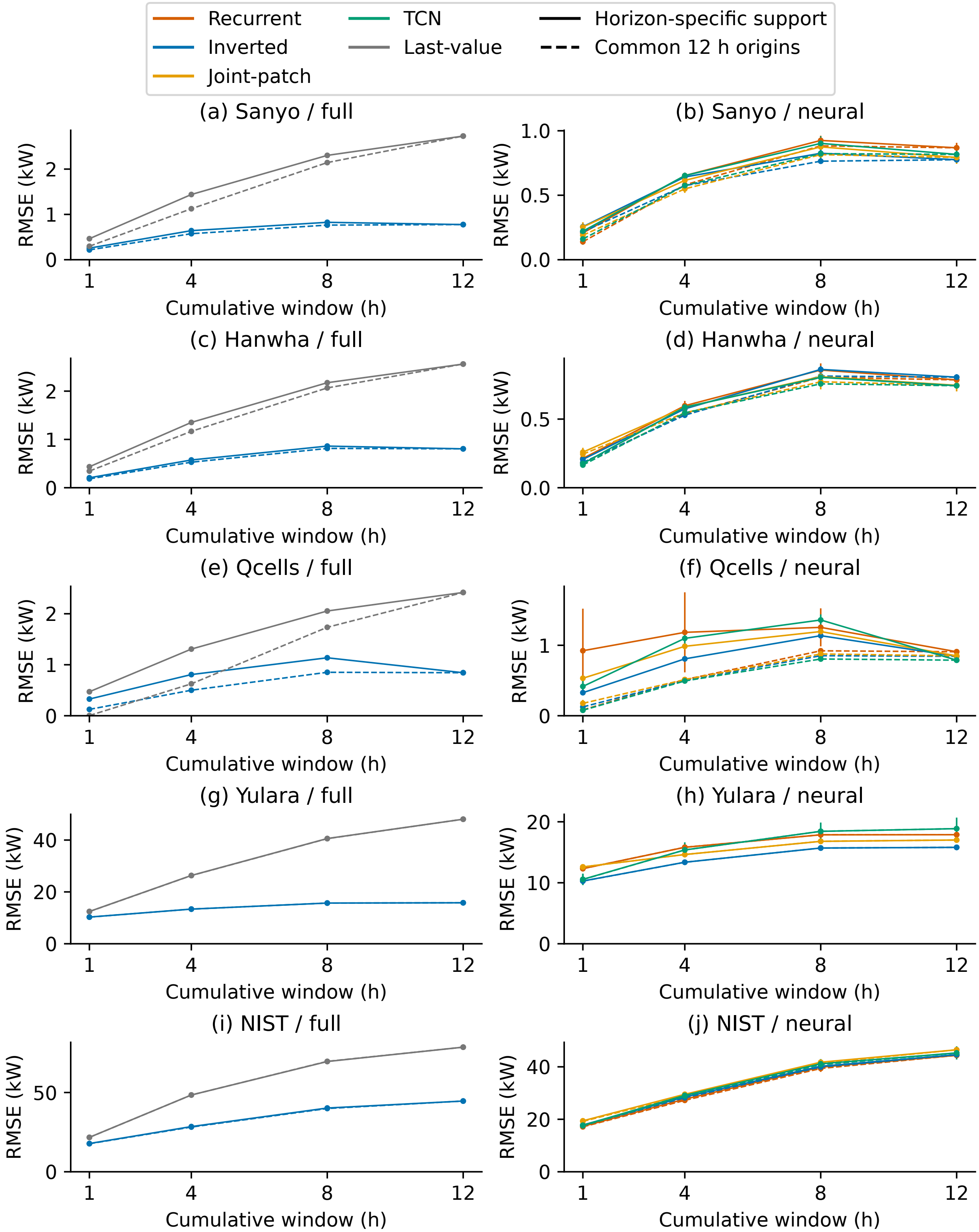

Figure S4: Horizon-specific support (solid) versus common complete-twelve-hour origins (dashed) cumulative errors. Bars are seed SD. Left panels preserve reference error ranges; right panels show neural detail on separately labelled axes.

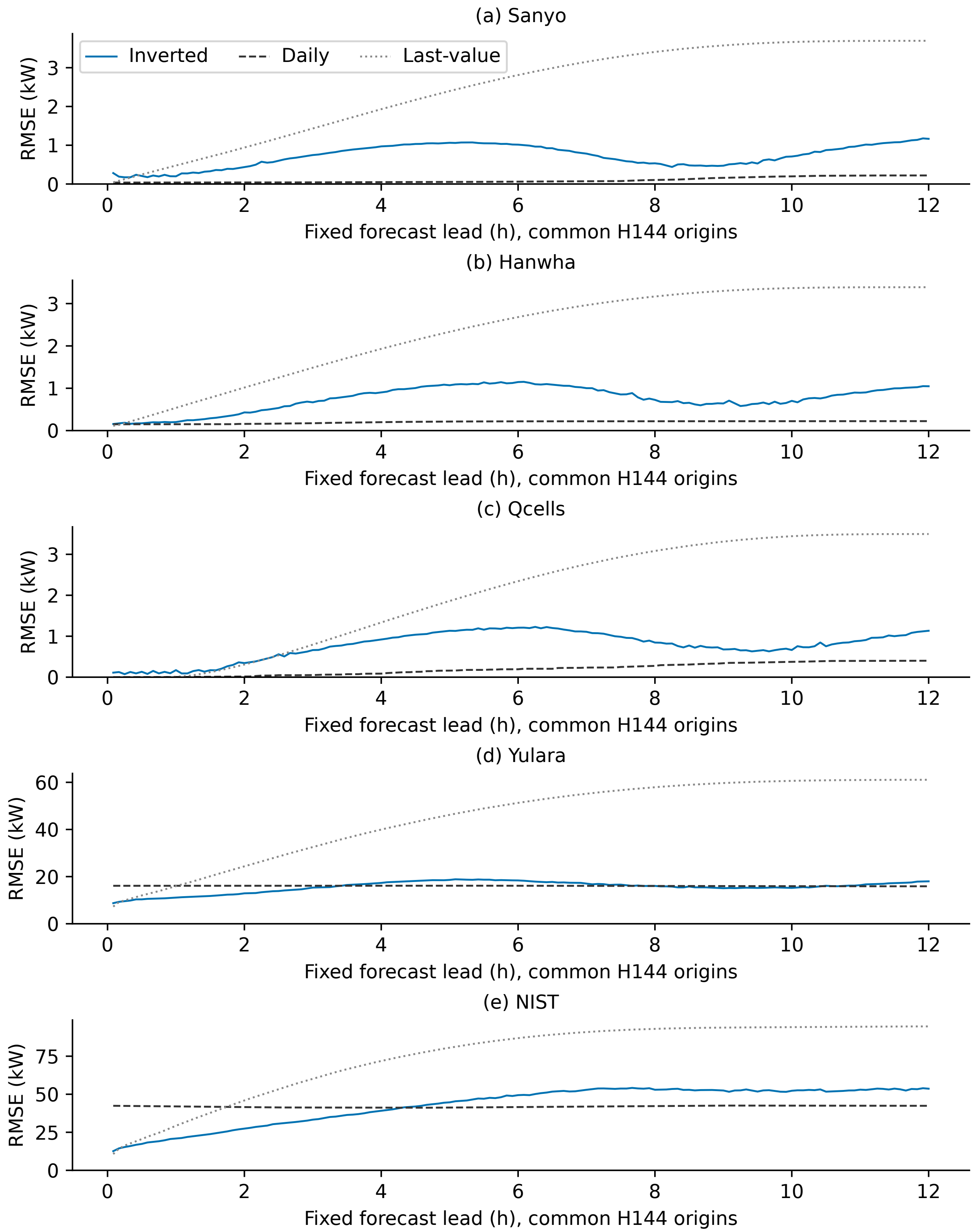

Figure S5: Lead-specific error at each five-minute lead on common complete-H144 origins and Daily-valid targets. These are not cumulative-prefix metrics. Neural curves average per-seed RMSE.

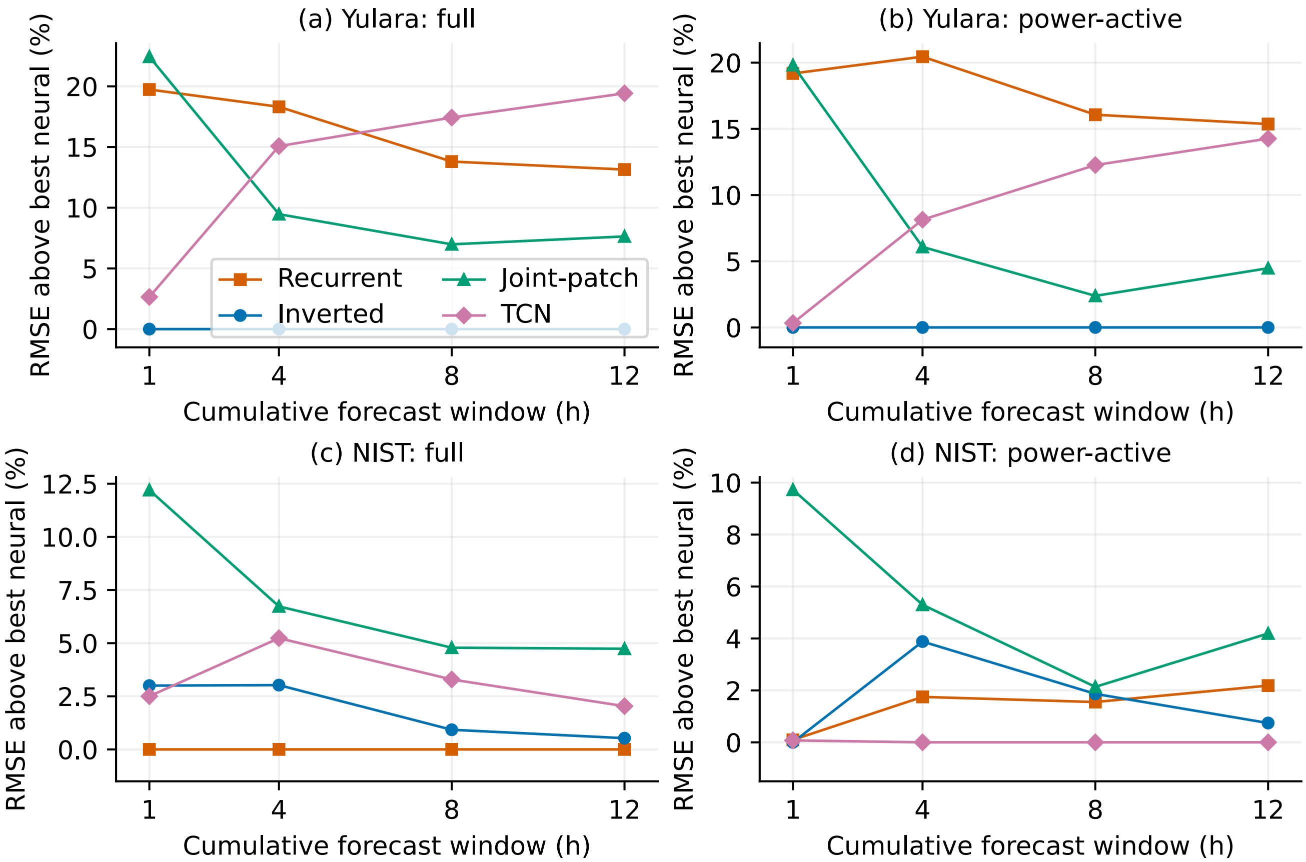

Figure S6: Relative RMSE gaps from the best mean neural implementation in each cumulative window and power scope. Zero identifies the lowest error; small gaps are shown as small gaps rather than enlarged rank differences.

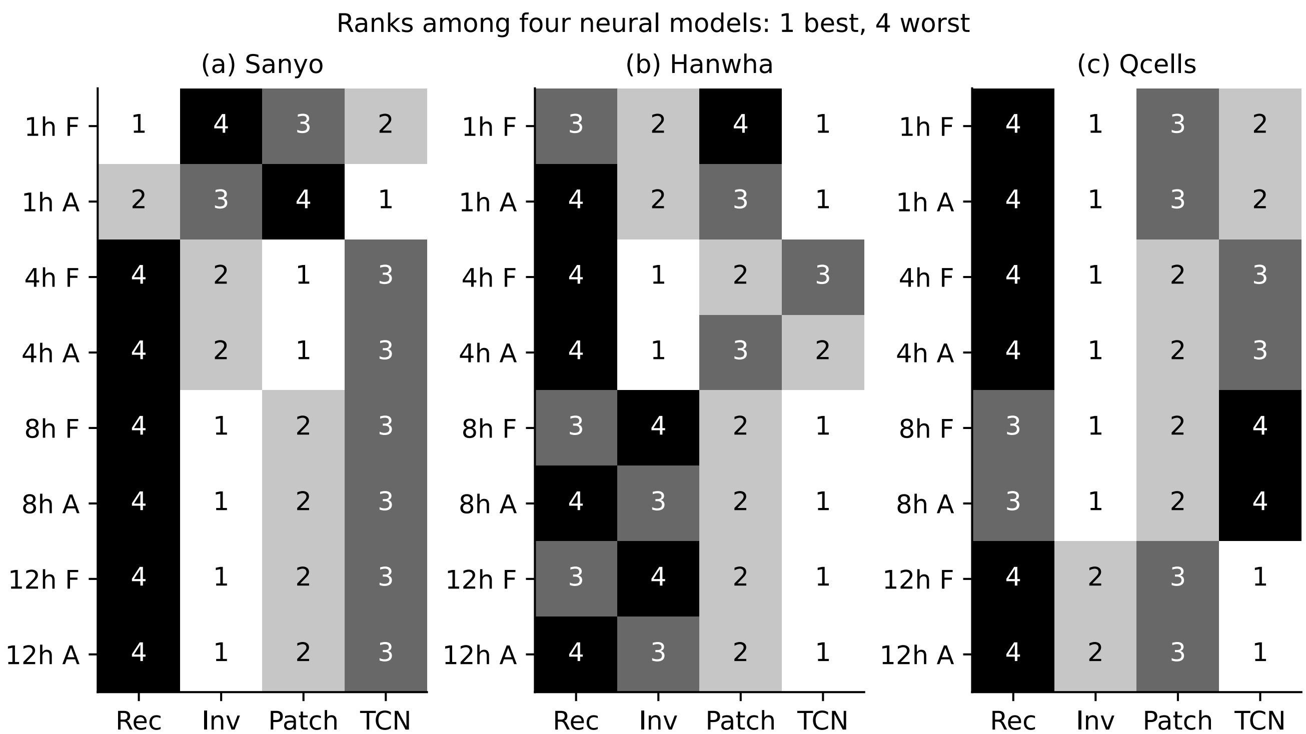

Figure S7: Alice neural ranks by cumulative window and target-power scope. F: full; A: power-active; four discrete rank levels.

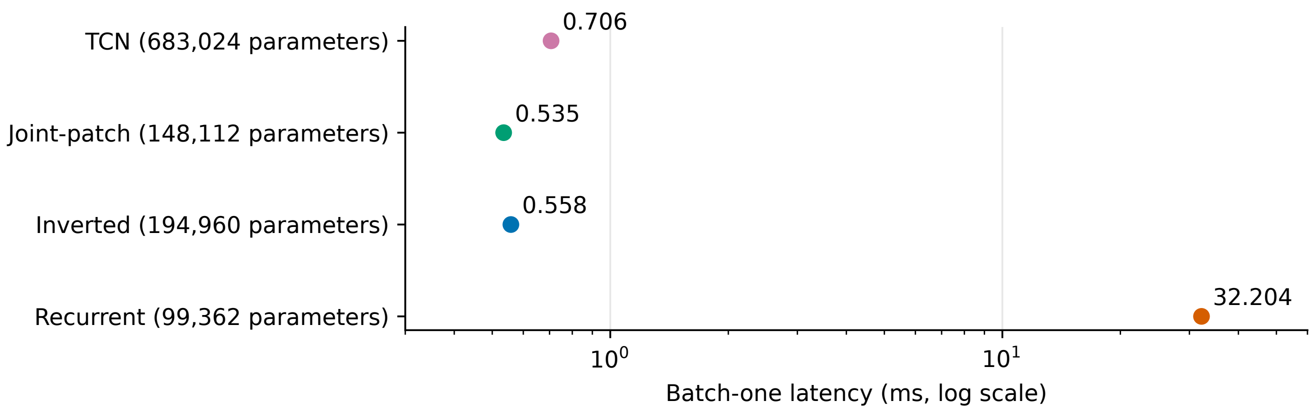

Figure S8: Historical 17-channel inference timings on RTX 3060 Laptop GPU, excluding loading/preprocessing. Point summaries do not establish a stable difference between 0.535 and 0.558 ms; no new timing experiment.

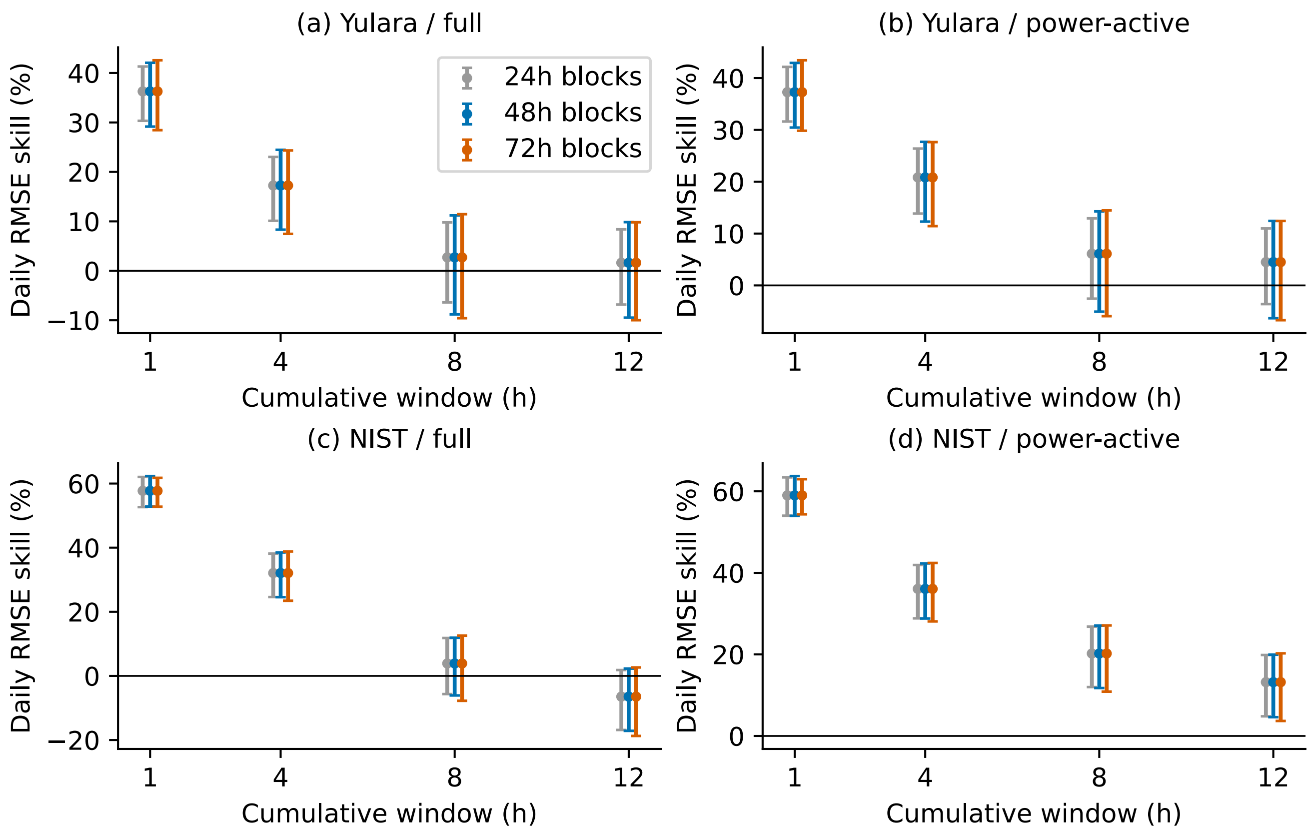

Figure S9: Paired temporal-block percentile intervals (2000 resamples; 24/48/72 h). SSE/count is aggregated before each seed RMSE, then seed skills are averaged. These intervals are conditional on observed dates and fitted models, not prediction intervals.

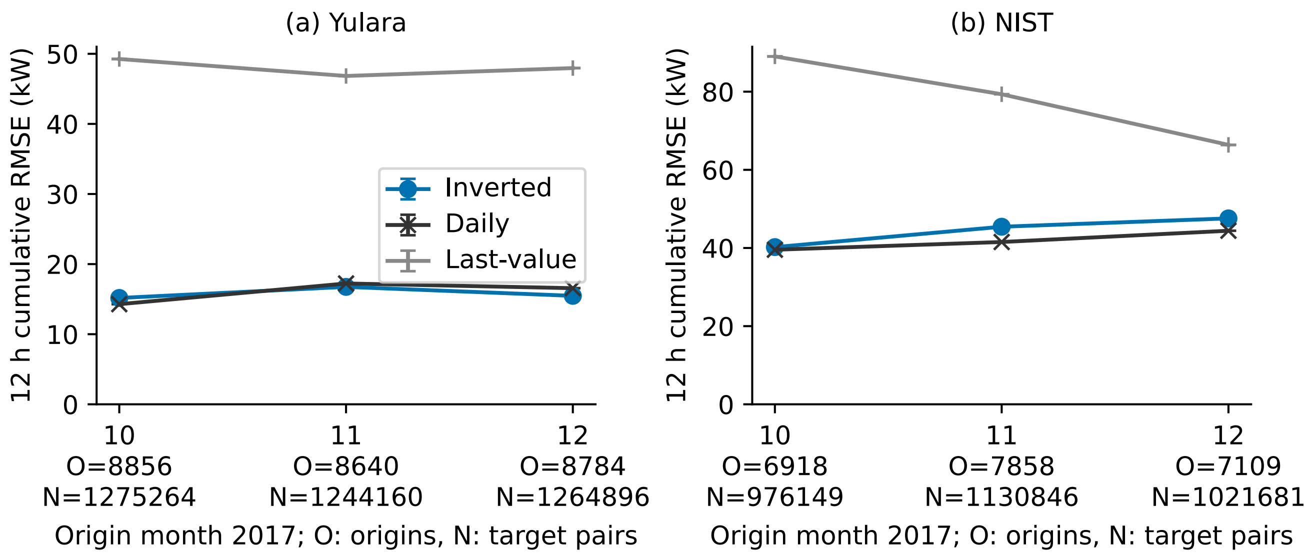

Figure S10: Monthly errors assign each full twelve-hour trajectory to its forecast-origin month on common H144 Daily-valid support. Bars show seed SD. O denotes origins and N scored origin–lead target pairs, printed for each month.

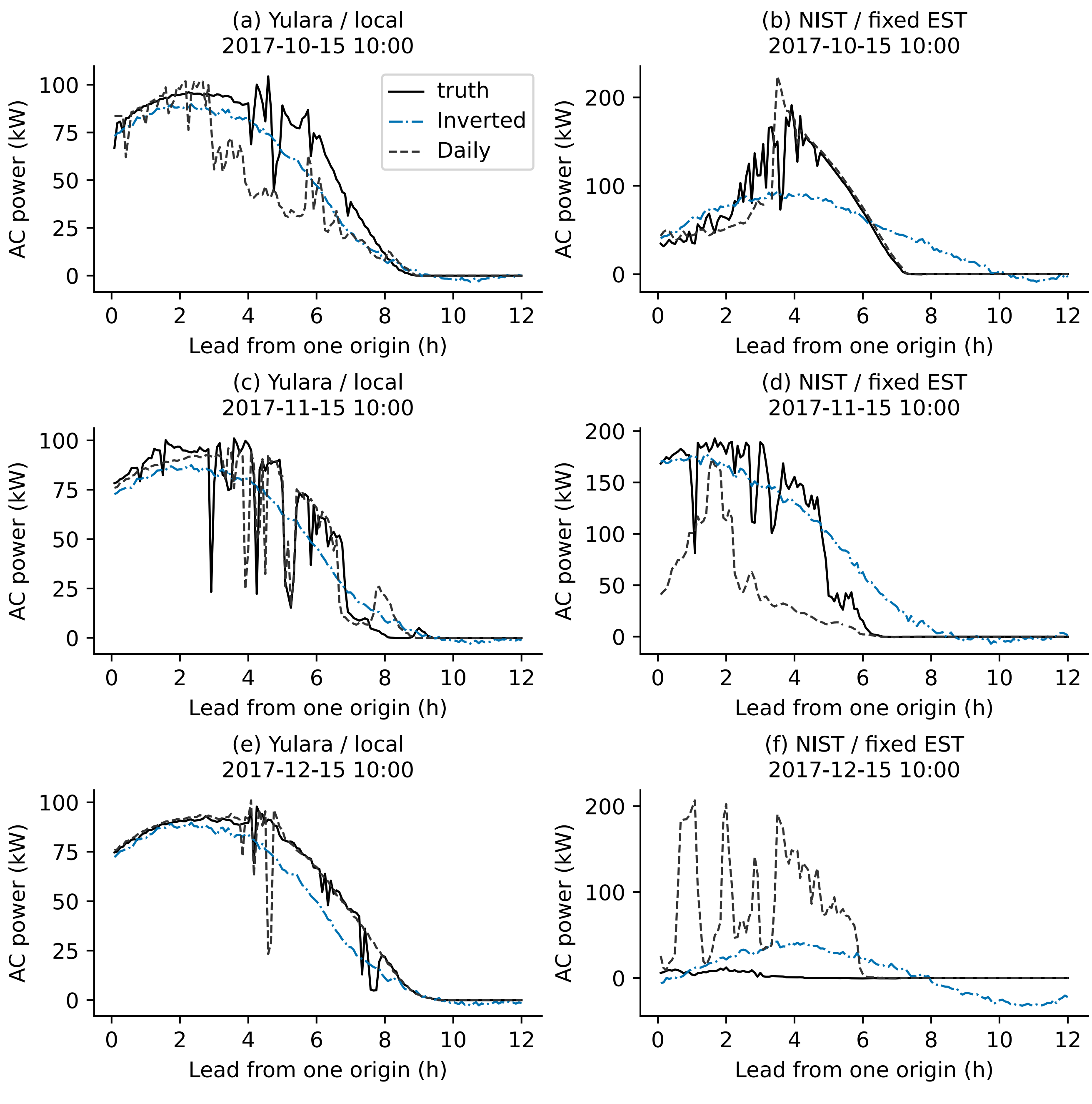

Figure S11: Fixed calendar seed42 trajectories. Fixed cases use the first eligible origin on or after each month's 15th at 10:00; extremes use original Daily SSE differences and are not representative frequencies.

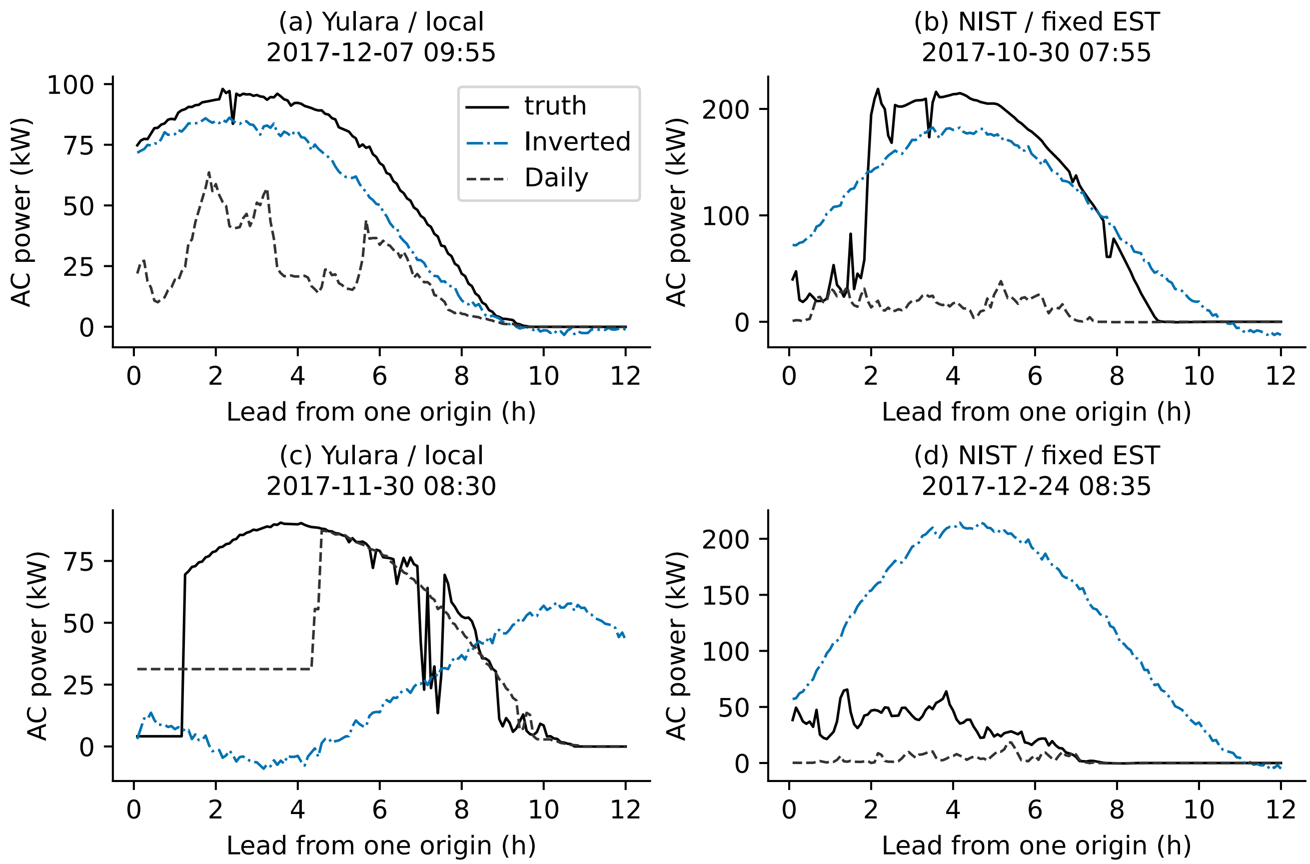

Figure S12: Post hoc diagnostic extremes seed42 trajectories. Fixed cases use the first eligible origin on or after each month's 15th at 10:00; extremes use original Daily SSE differences and are not representative frequencies.

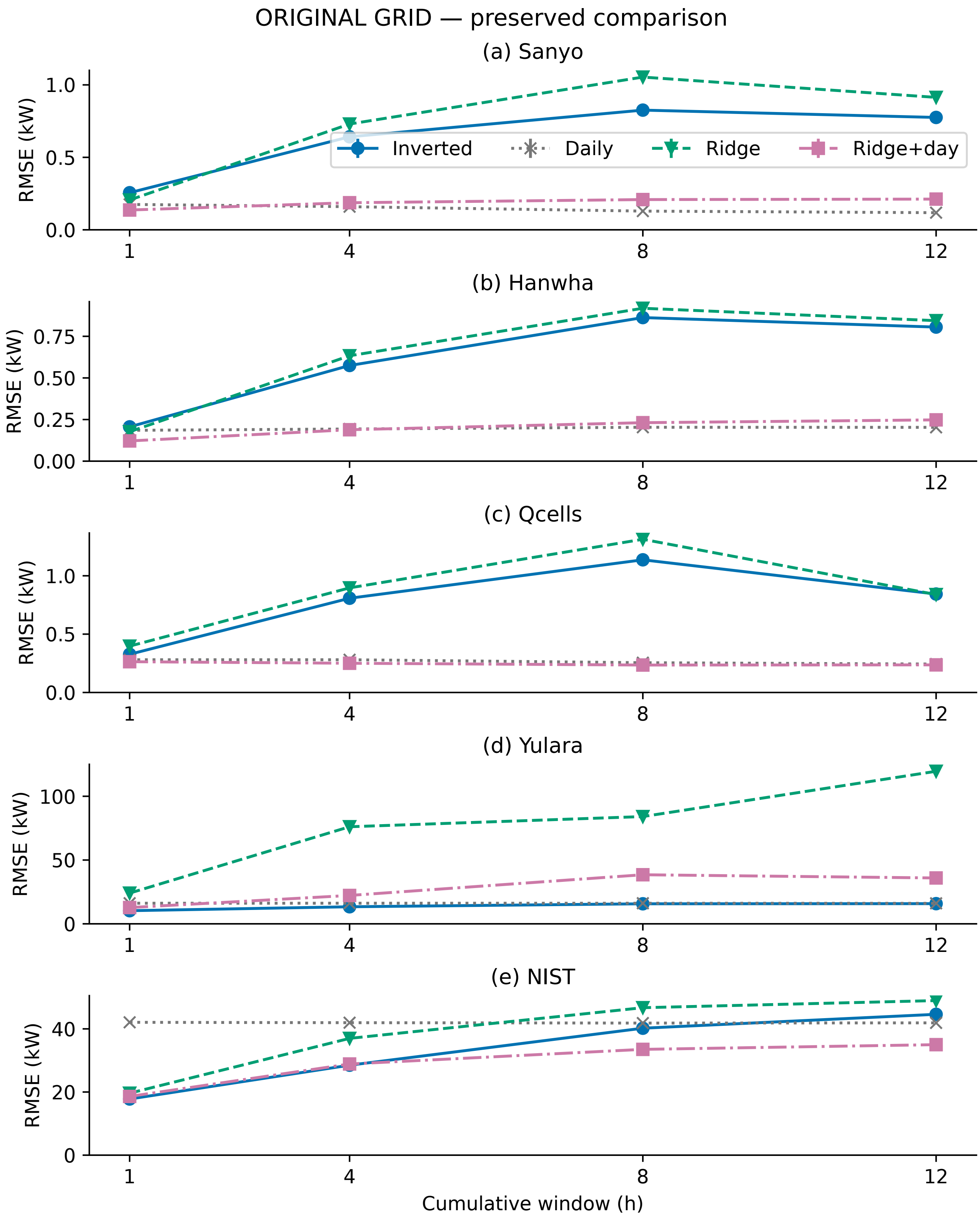

Figure S13: **Original grid.** Post hoc Ridge with recent history and Ridge plus available previous-day trajectory; alphas selected on Validation only. Daily-matched full targets are identical to the neural comparison. Ridge is deterministic; neural bars are three-seed SD.

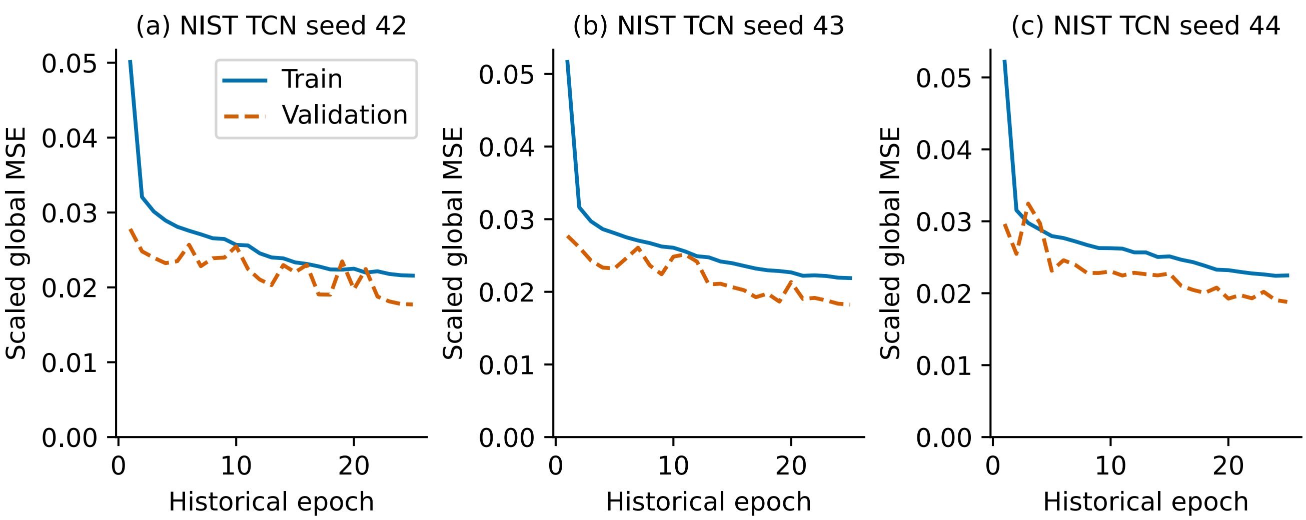

Figure S14: Historical NIST TCN learning records. All three selected checkpoints occur at the fixed maximum budget of 25 epochs. These curves document the budget boundary, not convergence or a new fit.

Historical benchmark definitions and complete tables

4 Evidence scope

This supplement reports the frozen evidence supporting the main manuscript. The benchmark comprises three co-located Alice Springs arrays (36 runs, 17 channels) and two separately fitted external facilities, Yulara and NIST Ground (24 runs, seven channels). Each experiment uses four compact neural implementations, seeds 42, 43, and 44, and 144-step trajectory forecasts at five-minute resolution. Primary H12, H48, H96, and H144 metrics use horizon-specific valid origins. Daily Persistence comparisons instead use an elementwise intersection mask that is valid for the daily lag, labels, Last-value Persistence, and every neural forecast.

5 Input and preprocessing configuration

For Alice Springs, each history has 72 time steps and 17 channels: Performance Ratio; ambient temperature; relative humidity; global and diffuse horizontal radiation; global and diffuse tilted radiation; Active Power; eight corresponding missingness indicators; and one Isolation Forest indicator. The imputer, feature scaler, target scaler, and Isolation Forest are fitted on Train only. No forecast-origin successor value enters the input. Windows remain within chronological splits and within a regular five-minute coordinate.

Table S1: Fixed optimization configuration for all compact neural implementations.
Alt text: The shared optimization configuration fixes the optimizer, learning rate, batch size, epoch budget, early-stopping rule, gradient clipping, H144 validation objective, and three random seeds for all four compact implementations.

Setting	Value
Optimizer	AdamW
Learning rate	10^{-3}
Weight decay	10^{-5}
Batch size	256
Maximum epochs	25
Early-stopping patience	5
Minimum improvement	10^{-8}
Gradient clipping	1.0
Checkpoint objective	Validation global masked MSE on H144
Random seeds	42, 43, 44

6 Architecture identity and full layer configuration

The Discrete recurrent decoder combines three temporal-convolution branches (dilations 1, 2, and 4; 24 channels each), a 64-dimensional one-layer Transformer, a learned feature gate, bidirectional GRU attention, and a recurrent hidden-state GRUCell trajectory decoder with fixed context and no previous-power feedback. The Inverted-variate Transformer treats the 17 Alice or seven external channels as tokens, projects each 72-point history to 64 dimensions, applies one four-head Transformer layer, then flattens the encoded tokens into a 144-output linear map. The Joint-patch Transformer forms overlapping joint multivariate patches of length 12 and stride 6, uses one 64-dimensional four-head Transformer layer, and maps the encoded patches to 144 outputs. The Depthwise TCN uses an input projection followed by four depthwise-separable temporal blocks with 64 channels and kernel size 5, then a flattened linear H144 head.

These are project implementations for a controlled mechanism comparison. The inverted model omits the full official iTransformer embedding, normalization, and decoder design; the joint-patch model does not use the channel-independent PatchTST pipeline; and the depthwise model omits the full ModernTCN large-kernel and multi-scale block system. The recurrent convolution branches use dropout 0.1; both compact attention encoders use the PyTorch TransformerEncoderLayer default dropout 0.1 and layer normalization. The TCN has neither an added dropout nor a normalization layer. Accordingly, the reported rankings concern the listed implementations.

7 Metrics and masks

For a fixed valid-point set $\mathcal{I}$ with $n = |\mathcal{I}|$,

$$\text{RMSE} = \sqrt{n^{-1} \sum_{i \in \mathcal{I}} (\hat{y}_i - y_i)^2}, \tag{S1}$$

$$\text{MAE} = n^{-1} \sum_{i \in \mathcal{I}} |\hat{y}_i - y_i|, \tag{S2}$$

$$R^2 = 1 - \frac{\sum_{i \in \mathcal{I}} (\hat{y}_i - y_i)^2}{\sum_{i \in \mathcal{I}} (y_i - \bar{y})^2}, \tag{S3}$$

$$\text{range-nRMSE(\%)} = 100 \frac{\text{RMSE}}{y_{\text{max,Train}} - y_{\text{min,Train}}}, \tag{S4}$$

$$\text{RMSE skill(\%)} = 100 \left(1 - \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{reference}}}\right). \tag{S5}$$

The power-active mask is true target Active Power greater than 1% of the corresponding array's Train maximum. It is an evaluation-only mask. Daily Persistence takes each target timestamp's power 24 hours earlier without interpolation. If that point is unavailable, it is removed from every method in that Daily-matched comparison.

8 Alice Springs quantitative evidence

The following tables are generated directly from the frozen `corrected_metrics.csv`; they are not transcribed into this source file. They provide per-seed primary metrics, primary support, complete-H144-origin sensitivity, and matched Daily Persistence results.

Table S2: Primary per-seed RMSE values.

Alt text: Per-seed root mean square errors show how forecast accuracy changes across arrays, scopes, and horizons for the four compact neural implementations. Seed spread reflects initialization variability rather than independent data replication.

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	1 h	Discrete recurrent	0.20699	0.21172	0.20282
Sanyo	Full	1 h	Inverted-variate	0.26513	0.27584	0.22716
Sanyo	Full	1 h	Joint-patch	0.29326	0.24817	0.21943
Sanyo	Full	1 h	Depthwise TCN	0.23350	0.17550	0.25439
Sanyo	Power-active	1 h	Discrete recurrent	0.29601	0.30675	0.29114
Sanyo	Power-active	1 h	Inverted-variate	0.31789	0.32546	0.30911
Sanyo	Power-active	1 h	Joint-patch	0.37815	0.32296	0.27677
Sanyo	Power-active	1 h	Depthwise TCN	0.31123	0.24041	0.32051
Sanyo	Full	4 h	Discrete recurrent	0.64683	0.64400	0.66629

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	4 h	Inverted-variate	0.68041	0.62303	0.62109
Sanyo	Full	4 h	Joint-patch	0.66859	0.58335	0.59367
Sanyo	Full	4 h	Depthwise TCN	0.67710	0.64485	0.63471
Sanyo	Power-active	4 h	Discrete recurrent	0.94975	0.94651	0.97978
Sanyo	Power-active	4 h	Inverted-variate	0.95042	0.81287	0.85809
Sanyo	Power-active	4 h	Joint-patch	0.95240	0.80115	0.83710
Sanyo	Power-active	4 h	Depthwise TCN	0.94231	0.90116	0.87112
Sanyo	Full	8 h	Discrete recurrent	0.90840	0.90011	0.96696
Sanyo	Full	8 h	Inverted-variate	0.85729	0.80883	0.81027
Sanyo	Full	8 h	Joint-patch	0.96456	0.84964	0.81070
Sanyo	Full	8 h	Depthwise TCN	0.96364	0.85757	0.88544
Sanyo	Power-active	8 h	Discrete recurrent	1.34860	1.33368	1.43813
Sanyo	Power-active	8 h	Inverted-variate	1.19297	1.08960	1.11853
Sanyo	Power-active	8 h	Joint-patch	1.39986	1.20066	1.14794
Sanyo	Power-active	8 h	Depthwise TCN	1.36564	1.19882	1.24128
Sanyo	Full	12 h	Discrete recurrent	0.87822	0.82591	0.89962
Sanyo	Full	12 h	Inverted-variate	0.78097	0.76385	0.78197
Sanyo	Full	12 h	Joint-patch	0.84637	0.78215	0.75005
Sanyo	Full	12 h	Depthwise TCN	0.85780	0.79629	0.79624
Sanyo	Power-active	12 h	Discrete recurrent	1.31199	1.22879	1.34119
Sanyo	Power-active	12 h	Inverted-variate	1.07005	1.02634	1.07535
Sanyo	Power-active	12 h	Joint-patch	1.22156	1.10516	1.06103
Sanyo	Power-active	12 h	Depthwise TCN	1.20759	1.10868	1.11171
Hanwha	Full	1 h	Discrete recurrent	0.19283	0.20984	0.23773
Hanwha	Full	1 h	Inverted-variate	0.19326	0.21233	0.21254
Hanwha	Full	1 h	Joint-patch	0.25045	0.29434	0.23248
Hanwha	Full	1 h	Depthwise TCN	0.17950	0.16762	0.17960
Hanwha	Power-active	1 h	Discrete recurrent	0.27411	0.30308	0.34010
Hanwha	Power-active	1 h	Inverted-variate	0.25729	0.28560	0.28441
Hanwha	Power-active	1 h	Joint-patch	0.31341	0.31390	0.28217
Hanwha	Power-active	1 h	Depthwise TCN	0.23343	0.22040	0.23794
Hanwha	Full	4 h	Discrete recurrent	0.57998	0.57396	0.63790
Hanwha	Full	4 h	Inverted-variate	0.54741	0.58955	0.58646
Hanwha	Full	4 h	Joint-patch	0.61852	0.57368	0.56605
Hanwha	Full	4 h	Depthwise TCN	0.57787	0.58575	0.59752
Hanwha	Power-active	4 h	Discrete recurrent	0.84718	0.84454	0.93710
Hanwha	Power-active	4 h	Inverted-variate	0.78356	0.84091	0.82613
Hanwha	Power-active	4 h	Joint-patch	0.88106	0.80086	0.79516
Hanwha	Power-active	4 h	Depthwise TCN	0.81332	0.82086	0.84039
Hanwha	Full	8 h	Discrete recurrent	0.80824	0.85950	0.90481
Hanwha	Full	8 h	Inverted-variate	0.84121	0.85535	0.88964
Hanwha	Full	8 h	Joint-patch	0.87878	0.76678	0.78162
Hanwha	Full	8 h	Depthwise TCN	0.79720	0.79868	0.81582
Hanwha	Power-active	8 h	Discrete recurrent	1.19164	1.27417	1.34138
Hanwha	Power-active	8 h	Inverted-variate	1.21196	1.22202	1.26041
Hanwha	Power-active	8 h	Joint-patch	1.27278	1.08927	1.11426
Hanwha	Power-active	8 h	Depthwise TCN	1.13261	1.12349	1.15324
Hanwha	Full	12 h	Discrete recurrent	0.76065	0.78426	0.81177
Hanwha	Full	12 h	Inverted-variate	0.78583	0.80309	0.82715

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Hanwha	Full	12 h	Joint-patch	0.79506	0.71160	0.73154
Hanwha	Full	12 h	Depthwise TCN	0.72555	0.75028	0.75361
Hanwha	Power-active	12 h	Discrete recurrent	1.12700	1.16621	1.20763
Hanwha	Power-active	12 h	Inverted-variate	1.12435	1.13795	1.16060
Hanwha	Power-active	12 h	Joint-patch	1.14106	1.00309	1.03867
Hanwha	Power-active	12 h	Depthwise TCN	1.01361	1.05240	1.05096
Qcells	Full	1 h	Discrete recurrent	0.44262	0.73507	1.59137
Qcells	Full	1 h	Inverted-variate	0.30176	0.34935	0.33091
Qcells	Full	1 h	Joint-patch	0.52988	0.57476	0.49208
Qcells	Full	1 h	Depthwise TCN	0.52598	0.30576	0.41648
Qcells	Power-active	1 h	Discrete recurrent	0.64247	1.06714	2.31558
Qcells	Power-active	1 h	Inverted-variate	0.42778	0.47506	0.46668
Qcells	Power-active	1 h	Joint-patch	0.72858	0.81902	0.69755
Qcells	Power-active	1 h	Depthwise TCN	0.75796	0.43460	0.60317
Qcells	Full	4 h	Discrete recurrent	0.77406	0.94518	1.83265
Qcells	Full	4 h	Inverted-variate	0.79118	0.82091	0.81042
Qcells	Full	4 h	Joint-patch	0.93780	1.03256	0.98688
Qcells	Full	4 h	Depthwise TCN	1.20653	1.04753	1.04156
Qcells	Power-active	4 h	Discrete recurrent	1.12663	1.37313	2.67354
Qcells	Power-active	4 h	Inverted-variate	1.14009	1.17767	1.16234
Qcells	Power-active	4 h	Joint-patch	1.33819	1.49374	1.42803
Qcells	Power-active	4 h	Depthwise TCN	1.75350	1.52143	1.51385
Qcells	Full	8 h	Discrete recurrent	1.04609	1.15831	1.55933
Qcells	Full	8 h	Inverted-variate	1.15002	1.14625	1.11365
Qcells	Full	8 h	Joint-patch	1.18191	1.23303	1.17040
Qcells	Full	8 h	Depthwise TCN	1.36166	1.43712	1.27932
Qcells	Power-active	8 h	Discrete recurrent	1.54353	1.70080	2.29733
Qcells	Power-active	8 h	Inverted-variate	1.67275	1.66186	1.60714
Qcells	Power-active	8 h	Joint-patch	1.71311	1.79827	1.70418
Qcells	Power-active	8 h	Depthwise TCN	1.99499	2.10755	1.86964
Qcells	Full	12 h	Discrete recurrent	0.90764	0.92812	0.88946
Qcells	Full	12 h	Inverted-variate	0.84633	0.86845	0.81567
Qcells	Full	12 h	Joint-patch	0.87026	0.85503	0.82492
Qcells	Full	12 h	Depthwise TCN	0.79967	0.79287	0.77389
Qcells	Power-active	12 h	Discrete recurrent	1.36729	1.38531	1.32341
Qcells	Power-active	12 h	Inverted-variate	1.23444	1.25633	1.17410
Qcells	Power-active	12 h	Joint-patch	1.26299	1.25253	1.20653
Qcells	Power-active	12 h	Depthwise TCN	1.17205	1.16000	1.12490

Table S3: Primary per-seed MAE values.

Alt text: Per-seed mean absolute errors provide a less tail-sensitive comparison of the four compact neural implementations across arrays, scopes, and horizons. The pattern complements RMSE by reducing the influence of large residuals.

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	1 h	Discrete recurrent	0.13583	0.12116	0.12189
Sanyo	Full	1 h	Inverted-variate	0.20606	0.20906	0.15958
Sanyo	Full	1 h	Joint-patch	0.21315	0.17464	0.16188

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	1 h	Depthwise TCN	0.17238	0.11426	0.20034
Sanyo	Power-active	1 h	Discrete recurrent	0.22632	0.21644	0.20251
Sanyo	Power-active	1 h	Inverted-variate	0.24672	0.25630	0.24797
Sanyo	Power-active	1 h	Joint-patch	0.29948	0.25203	0.21011
Sanyo	Power-active	1 h	Depthwise TCN	0.25914	0.17448	0.25712
Sanyo	Full	4 h	Discrete recurrent	0.31052	0.26504	0.28877
Sanyo	Full	4 h	Inverted-variate	0.41494	0.40296	0.36613
Sanyo	Full	4 h	Joint-patch	0.37397	0.33657	0.32167
Sanyo	Full	4 h	Depthwise TCN	0.40982	0.37110	0.37789
Sanyo	Power-active	4 h	Discrete recurrent	0.59637	0.52821	0.55362
Sanyo	Power-active	4 h	Inverted-variate	0.63958	0.52668	0.58159
Sanyo	Power-active	4 h	Joint-patch	0.61831	0.49596	0.51232
Sanyo	Power-active	4 h	Depthwise TCN	0.64092	0.58876	0.54797
Sanyo	Full	8 h	Discrete recurrent	0.45539	0.42600	0.46402
Sanyo	Full	8 h	Inverted-variate	0.54810	0.53406	0.50433
Sanyo	Full	8 h	Joint-patch	0.55816	0.51419	0.46833
Sanyo	Full	8 h	Depthwise TCN	0.61608	0.52564	0.54538
Sanyo	Power-active	8 h	Discrete recurrent	0.92720	0.88733	0.95360
Sanyo	Power-active	8 h	Inverted-variate	0.88941	0.80380	0.83193
Sanyo	Power-active	8 h	Joint-patch	1.00084	0.85190	0.79454
Sanyo	Power-active	8 h	Depthwise TCN	1.05117	0.88114	0.89613
Sanyo	Full	12 h	Discrete recurrent	0.45801	0.40717	0.46249
Sanyo	Full	12 h	Inverted-variate	0.50741	0.51151	0.50399
Sanyo	Full	12 h	Joint-patch	0.49204	0.47784	0.44669
Sanyo	Full	12 h	Depthwise TCN	0.56495	0.51928	0.51794
Sanyo	Power-active	12 h	Discrete recurrent	0.96352	0.86876	0.97403
Sanyo	Power-active	12 h	Inverted-variate	0.81015	0.78388	0.83433
Sanyo	Power-active	12 h	Joint-patch	0.87534	0.80142	0.76574
Sanyo	Power-active	12 h	Depthwise TCN	0.95405	0.87339	0.86283
Hanwha	Full	1 h	Discrete recurrent	0.12387	0.12836	0.16202
Hanwha	Full	1 h	Inverted-variate	0.13972	0.15415	0.15395
Hanwha	Full	1 h	Joint-patch	0.19486	0.24511	0.17962
Hanwha	Full	1 h	Depthwise TCN	0.12813	0.11327	0.12656
Hanwha	Power-active	1 h	Discrete recurrent	0.20778	0.22419	0.28126
Hanwha	Power-active	1 h	Inverted-variate	0.19702	0.22770	0.22425
Hanwha	Power-active	1 h	Joint-patch	0.25158	0.24992	0.22135
Hanwha	Power-active	1 h	Depthwise TCN	0.18270	0.15758	0.18161
Hanwha	Full	4 h	Discrete recurrent	0.27710	0.23620	0.32083
Hanwha	Full	4 h	Inverted-variate	0.28438	0.32693	0.33336
Hanwha	Full	4 h	Joint-patch	0.36652	0.34526	0.33407
Hanwha	Full	4 h	Depthwise TCN	0.33142	0.33673	0.34221
Hanwha	Power-active	4 h	Discrete recurrent	0.52195	0.46421	0.62989
Hanwha	Power-active	4 h	Inverted-variate	0.47541	0.54087	0.52920
Hanwha	Power-active	4 h	Joint-patch	0.59312	0.52110	0.52779
Hanwha	Power-active	4 h	Depthwise TCN	0.52569	0.53522	0.53927
Hanwha	Full	8 h	Discrete recurrent	0.40774	0.40206	0.45383
Hanwha	Full	8 h	Inverted-variate	0.49253	0.52614	0.55511
Hanwha	Full	8 h	Joint-patch	0.52264	0.46726	0.46749
Hanwha	Full	8 h	Depthwise TCN	0.47434	0.48192	0.49329

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Hanwha	Power-active	8 h	Discrete recurrent	0.81616	0.83369	0.93289
Hanwha	Power-active	8 h	Inverted-variate	0.87000	0.91131	0.93783
Hanwha	Power-active	8 h	Joint-patch	0.92079	0.77157	0.79984
Hanwha	Power-active	8 h	Depthwise TCN	0.81254	0.81988	0.83932
Hanwha	Full	12 h	Discrete recurrent	0.40071	0.38910	0.42640
Hanwha	Full	12 h	Inverted-variate	0.48388	0.52199	0.53621
Hanwha	Full	12 h	Joint-patch	0.47310	0.44847	0.44631
Hanwha	Full	12 h	Depthwise TCN	0.44980	0.47854	0.47476
Hanwha	Power-active	12 h	Discrete recurrent	0.81786	0.82031	0.89037
Hanwha	Power-active	12 h	Inverted-variate	0.84028	0.88713	0.89231
Hanwha	Power-active	12 h	Joint-patch	0.82794	0.73477	0.76446
Hanwha	Power-active	12 h	Depthwise TCN	0.74890	0.81601	0.79182
Qcells	Full	1 h	Discrete recurrent	0.26715	0.44107	0.92759
Qcells	Full	1 h	Inverted-variate	0.17351	0.23238	0.20505
Qcells	Full	1 h	Joint-patch	0.38756	0.38343	0.33033
Qcells	Full	1 h	Depthwise TCN	0.33873	0.18634	0.23193
Qcells	Power-active	1 h	Discrete recurrent	0.51963	0.87320	1.85785
Qcells	Power-active	1 h	Inverted-variate	0.28354	0.34031	0.33801
Qcells	Power-active	1 h	Joint-patch	0.59609	0.67195	0.56727
Qcells	Power-active	1 h	Depthwise TCN	0.62824	0.31843	0.44563
Qcells	Full	4 h	Discrete recurrent	0.38057	0.54036	1.10494
Qcells	Full	4 h	Inverted-variate	0.42459	0.46561	0.44915
Qcells	Full	4 h	Joint-patch	0.60764	0.63207	0.60534
Qcells	Full	4 h	Depthwise TCN	0.70893	0.59737	0.57099
Qcells	Power-active	4 h	Discrete recurrent	0.75199	1.05716	2.25583
Qcells	Power-active	4 h	Inverted-variate	0.77200	0.81193	0.79177
Qcells	Power-active	4 h	Joint-patch	1.03945	1.18366	1.13374
Qcells	Power-active	4 h	Depthwise TCN	1.38908	1.15015	1.12203
Qcells	Full	8 h	Discrete recurrent	0.54199	0.65278	0.90353
Qcells	Full	8 h	Inverted-variate	0.66567	0.68290	0.66126
Qcells	Full	8 h	Joint-patch	0.74340	0.75035	0.71193
Qcells	Full	8 h	Depthwise TCN	0.78189	0.82302	0.73178
Qcells	Power-active	8 h	Discrete recurrent	1.11654	1.31308	1.84620
Qcells	Power-active	8 h	Inverted-variate	1.24135	1.24596	1.19846
Qcells	Power-active	8 h	Joint-patch	1.33358	1.41995	1.33442
Qcells	Power-active	8 h	Depthwise TCN	1.52256	1.61576	1.41140
Qcells	Full	12 h	Discrete recurrent	0.49640	0.51181	0.48935
Qcells	Full	12 h	Inverted-variate	0.52191	0.56507	0.51260
Qcells	Full	12 h	Joint-patch	0.56861	0.53326	0.50553
Qcells	Full	12 h	Depthwise TCN	0.49955	0.47670	0.47809
Qcells	Power-active	12 h	Discrete recurrent	1.06417	1.04785	0.97894
Qcells	Power-active	12 h	Inverted-variate	0.94392	0.98971	0.90037
Qcells	Power-active	12 h	Joint-patch	0.97849	0.97913	0.92040
Qcells	Power-active	12 h	Depthwise TCN	0.93775	0.88304	0.86744

Table S4: Primary per-seed R2 values.

Alt text: Per-seed coefficients of determination show horizon- and scope-dependent explained variance, including negative values where forecasts underperform the target mean. Values are descriptive because adjacent forecast origins overlap.

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	1 h	Discrete recurrent	0.99012	0.98966	0.99051
Sanyo	Full	1 h	Inverted-variate	0.98379	0.98246	0.98810
Sanyo	Full	1 h	Joint-patch	0.98017	0.98580	0.98890
Sanyo	Full	1 h	Depthwise TCN	0.98743	0.99290	0.98508
Sanyo	Power-active	1 h	Discrete recurrent	0.96350	0.96081	0.96469
Sanyo	Power-active	1 h	Inverted-variate	0.95791	0.95588	0.96020
Sanyo	Power-active	1 h	Joint-patch	0.94043	0.95655	0.96809
Sanyo	Power-active	1 h	Depthwise TCN	0.95965	0.97592	0.95721
Sanyo	Full	4 h	Discrete recurrent	0.90861	0.90941	0.90303
Sanyo	Full	4 h	Inverted-variate	0.89887	0.91521	0.91574
Sanyo	Full	4 h	Joint-patch	0.90236	0.92567	0.92301
Sanyo	Full	4 h	Depthwise TCN	0.89986	0.90917	0.91200
Sanyo	Power-active	4 h	Discrete recurrent	0.59798	0.60072	0.57216
Sanyo	Power-active	4 h	Inverted-variate	0.59742	0.70551	0.67183
Sanyo	Power-active	4 h	Joint-patch	0.59573	0.71394	0.68769
Sanyo	Power-active	4 h	Depthwise TCN	0.60426	0.63806	0.66179
Sanyo	Full	8 h	Discrete recurrent	0.81383	0.81722	0.78906
Sanyo	Full	8 h	Inverted-variate	0.83419	0.85241	0.85188
Sanyo	Full	8 h	Joint-patch	0.79010	0.83714	0.85173
Sanyo	Full	8 h	Depthwise TCN	0.79050	0.83409	0.82312
Sanyo	Power-active	8 h	Discrete recurrent	0.22692	0.24393	0.12087
Sanyo	Power-active	8 h	Inverted-variate	0.39505	0.49534	0.46819
Sanyo	Power-active	8 h	Joint-patch	0.16703	0.38723	0.43986
Sanyo	Power-active	8 h	Depthwise TCN	0.20727	0.38910	0.34507
Sanyo	Full	12 h	Discrete recurrent	0.81869	0.83965	0.80974
Sanyo	Full	12 h	Inverted-variate	0.85662	0.86284	0.85625
Sanyo	Full	12 h	Joint-patch	0.83160	0.85618	0.86775
Sanyo	Full	12 h	Depthwise TCN	0.82702	0.85094	0.85096
Sanyo	Power-active	12 h	Discrete recurrent	0.30739	0.39245	0.27621
Sanyo	Power-active	12 h	Inverted-variate	0.53928	0.57615	0.53471
Sanyo	Power-active	12 h	Joint-patch	0.39957	0.50855	0.54701
Sanyo	Power-active	12 h	Depthwise TCN	0.41323	0.50541	0.50270
Hanwha	Full	1 h	Discrete recurrent	0.98994	0.98809	0.98471
Hanwha	Full	1 h	Inverted-variate	0.98989	0.98780	0.98778
Hanwha	Full	1 h	Joint-patch	0.98303	0.97656	0.98538
Hanwha	Full	1 h	Depthwise TCN	0.99128	0.99240	0.99127
Hanwha	Power-active	1 h	Discrete recurrent	0.96479	0.95696	0.94580
Hanwha	Power-active	1 h	Inverted-variate	0.96898	0.96178	0.96210
Hanwha	Power-active	1 h	Joint-patch	0.95397	0.95383	0.96269
Hanwha	Power-active	1 h	Depthwise TCN	0.97447	0.97724	0.97347
Hanwha	Full	4 h	Discrete recurrent	0.91295	0.91474	0.89469
Hanwha	Full	4 h	Inverted-variate	0.92245	0.91005	0.91099
Hanwha	Full	4 h	Joint-patch	0.90099	0.91483	0.91708
Hanwha	Full	4 h	Depthwise TCN	0.91358	0.91121	0.90760

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Hanwha	Power-active	4 h	Discrete recurrent	0.65097	0.65314	0.57294
Hanwha	Power-active	4 h	Inverted-variate	0.70142	0.65611	0.66810
Hanwha	Power-active	4 h	Joint-patch	0.62249	0.68809	0.69251
Hanwha	Power-active	4 h	Depthwise TCN	0.67831	0.67232	0.65654
Hanwha	Full	8 h	Discrete recurrent	0.82583	0.80303	0.78172
Hanwha	Full	8 h	Inverted-variate	0.81133	0.80494	0.78898
Hanwha	Full	8 h	Joint-patch	0.79410	0.84324	0.83711
Hanwha	Full	8 h	Depthwise TCN	0.83056	0.82993	0.82255
Hanwha	Power-active	8 h	Discrete recurrent	0.33672	0.24166	0.15955
Hanwha	Power-active	8 h	Inverted-variate	0.31390	0.30246	0.25794
Hanwha	Power-active	8 h	Joint-patch	0.24331	0.44578	0.42006
Hanwha	Power-active	8 h	Depthwise TCN	0.40080	0.41041	0.37877
Hanwha	Full	12 h	Discrete recurrent	0.84051	0.83045	0.81835
Hanwha	Full	12 h	Inverted-variate	0.82978	0.82222	0.81141
Hanwha	Full	12 h	Joint-patch	0.82575	0.86041	0.85248
Hanwha	Full	12 h	Depthwise TCN	0.85489	0.84483	0.84345
Hanwha	Power-active	12 h	Discrete recurrent	0.43177	0.39155	0.34756
Hanwha	Power-active	12 h	Inverted-variate	0.43445	0.42068	0.39739
Hanwha	Power-active	12 h	Joint-patch	0.41750	0.54985	0.51736
Hanwha	Power-active	12 h	Depthwise TCN	0.54036	0.50451	0.50586
Qcells	Full	1 h	Discrete recurrent	0.94988	0.86178	0.35216
Qcells	Full	1 h	Inverted-variate	0.97671	0.96878	0.97199
Qcells	Full	1 h	Joint-patch	0.92817	0.91549	0.93806
Qcells	Full	1 h	Depthwise TCN	0.92923	0.97608	0.95563
Qcells	Power-active	1 h	Discrete recurrent	0.81396	0.48674	-1.41663
Qcells	Power-active	1 h	Inverted-variate	0.91752	0.89828	0.90184
Qcells	Power-active	1 h	Joint-patch	0.76075	0.69767	0.78069
Qcells	Power-active	1 h	Depthwise TCN	0.74107	0.91487	0.83603
Qcells	Full	4 h	Discrete recurrent	0.85828	0.78869	0.20558
Qcells	Full	4 h	Inverted-variate	0.85194	0.84060	0.84465
Qcells	Full	4 h	Joint-patch	0.79198	0.74781	0.76963
Qcells	Full	4 h	Depthwise TCN	0.65567	0.74045	0.74340
Qcells	Power-active	4 h	Discrete recurrent	0.37554	0.07240	-2.51653
Qcells	Power-active	4 h	Inverted-variate	0.36053	0.31768	0.33533
Qcells	Power-active	4 h	Joint-patch	0.11900	-0.09771	-0.00327
Qcells	Power-active	4 h	Depthwise TCN	-0.51269	-0.13879	-0.12748
Qcells	Full	8 h	Discrete recurrent	0.72851	0.66714	0.39676
Qcells	Full	8 h	Inverted-variate	0.67189	0.67404	0.69231
Qcells	Full	8 h	Joint-patch	0.65344	0.62281	0.66015
Qcells	Full	8 h	Depthwise TCN	0.54001	0.48761	0.59396
Qcells	Power-active	8 h	Discrete recurrent	-0.08402	-0.31618	-1.40133
Qcells	Power-active	8 h	Inverted-variate	-0.27312	-0.25659	-0.17520
Qcells	Power-active	8 h	Joint-patch	-0.33529	-0.47135	-0.32141
Qcells	Power-active	8 h	Depthwise TCN	-0.81087	-1.02099	-0.59047
Qcells	Full	12 h	Discrete recurrent	0.78078	0.77078	0.78948
Qcells	Full	12 h	Inverted-variate	0.80940	0.79931	0.82296
Qcells	Full	12 h	Joint-patch	0.79847	0.80546	0.81892
Qcells	Full	12 h	Depthwise TCN	0.82984	0.83272	0.84063
Qcells	Power-active	12 h	Discrete recurrent	0.22854	0.20807	0.27726

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Qcells	Power-active	12 h	Inverted-variate	0.37117	0.34867	0.43114
Qcells	Power-active	12 h	Joint-patch	0.34174	0.35260	0.39928
Qcells	Power-active	12 h	Depthwise TCN	0.43312	0.44472	0.47782

Table S5: Primary per-seed Bias values.

Alt text: Per-seed mean errors identify underprediction and overprediction patterns across the three arrays, two scopes, and four forecast horizons. Positive values indicate overprediction, and negative values indicate underprediction.

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	1 h	Discrete recurrent	-0.01485	0.02522	-0.01926
Sanyo	Full	1 h	Inverted-variate	0.06001	0.08298	-0.00450
Sanyo	Full	1 h	Joint-patch	0.05334	0.07973	0.05282
Sanyo	Full	1 h	Depthwise TCN	-0.06692	0.03892	0.16604
Sanyo	Power-active	1 h	Discrete recurrent	-0.09490	0.04305	-0.06442
Sanyo	Power-active	1 h	Inverted-variate	-0.06615	0.02881	-0.02995
Sanyo	Power-active	1 h	Joint-patch	0.08527	0.10689	0.06451
Sanyo	Power-active	1 h	Depthwise TCN	-0.22977	0.06247	0.18508
Sanyo	Full	4 h	Discrete recurrent	-0.19674	-0.14171	-0.18156
Sanyo	Full	4 h	Inverted-variate	-0.15590	0.00905	-0.13983
Sanyo	Full	4 h	Joint-patch	-0.19055	-0.04334	-0.12676
Sanyo	Full	4 h	Depthwise TCN	-0.19644	-0.14263	-0.07699
Sanyo	Power-active	4 h	Discrete recurrent	-0.49498	-0.33281	-0.45605
Sanyo	Power-active	4 h	Inverted-variate	-0.54239	-0.31128	-0.46493
Sanyo	Power-active	4 h	Joint-patch	-0.44354	-0.25136	-0.35105
Sanyo	Power-active	4 h	Depthwise TCN	-0.58612	-0.43602	-0.37384
Sanyo	Full	8 h	Discrete recurrent	-0.33184	-0.30250	-0.35482
Sanyo	Full	8 h	Inverted-variate	-0.21373	-0.09706	-0.20703
Sanyo	Full	8 h	Joint-patch	-0.36865	-0.22148	-0.22546
Sanyo	Full	8 h	Depthwise TCN	-0.35401	-0.19358	-0.20143
Sanyo	Power-active	8 h	Discrete recurrent	-0.80950	-0.71352	-0.84974
Sanyo	Power-active	8 h	Inverted-variate	-0.66186	-0.54659	-0.61488
Sanyo	Power-active	8 h	Joint-patch	-0.84300	-0.59778	-0.54960
Sanyo	Power-active	8 h	Depthwise TCN	-0.90937	-0.60478	-0.67695
Sanyo	Full	12 h	Discrete recurrent	-0.30150	-0.24728	-0.28443
Sanyo	Full	12 h	Inverted-variate	-0.12415	-0.07899	-0.19460
Sanyo	Full	12 h	Joint-patch	-0.22895	-0.16462	-0.18296
Sanyo	Full	12 h	Depthwise TCN	-0.25089	-0.17817	-0.16368
Sanyo	Power-active	12 h	Discrete recurrent	-0.76490	-0.61458	-0.72355
Sanyo	Power-active	12 h	Inverted-variate	-0.45553	-0.45708	-0.55347
Sanyo	Power-active	12 h	Joint-patch	-0.57689	-0.49772	-0.44642
Sanyo	Power-active	12 h	Depthwise TCN	-0.70901	-0.59856	-0.57486
Hanwha	Full	1 h	Discrete recurrent	-0.07223	-0.06327	-0.12246
Hanwha	Full	1 h	Inverted-variate	-0.00845	-0.01508	-0.01574
Hanwha	Full	1 h	Joint-patch	0.09252	-0.19886	-0.05599
Hanwha	Full	1 h	Depthwise TCN	-0.02420	0.01298	0.03659
Hanwha	Power-active	1 h	Discrete recurrent	-0.13923	-0.12002	-0.21837
Hanwha	Power-active	1 h	Inverted-variate	0.00506	-0.08452	-0.07401

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Hanwha	Power-active	1 h	Joint-patch	0.06763	-0.15859	0.00038
Hanwha	Power-active	1 h	Depthwise TCN	-0.03043	-0.00929	0.05944
Hanwha	Full	4 h	Discrete recurrent	-0.21430	-0.17656	-0.27482
Hanwha	Full	4 h	Inverted-variate	-0.13472	-0.18278	-0.14005
Hanwha	Full	4 h	Joint-patch	-0.15171	-0.23340	-0.19411
Hanwha	Full	4 h	Depthwise TCN	-0.19526	-0.16190	-0.18508
Hanwha	Power-active	4 h	Discrete recurrent	-0.46980	-0.38290	-0.56927
Hanwha	Power-active	4 h	Inverted-variate	-0.35116	-0.44743	-0.40447
Hanwha	Power-active	4 h	Joint-patch	-0.45830	-0.40349	-0.37726
Hanwha	Power-active	4 h	Depthwise TCN	-0.41941	-0.45174	-0.41057
Hanwha	Full	8 h	Discrete recurrent	-0.32869	-0.33349	-0.39879
Hanwha	Full	8 h	Inverted-variate	-0.32324	-0.34001	-0.34498
Hanwha	Full	8 h	Joint-patch	-0.33245	-0.29819	-0.28419
Hanwha	Full	8 h	Depthwise TCN	-0.26653	-0.22095	-0.26397
Hanwha	Power-active	8 h	Discrete recurrent	-0.72991	-0.74335	-0.86099
Hanwha	Power-active	8 h	Inverted-variate	-0.72901	-0.76600	-0.78191
Hanwha	Power-active	8 h	Joint-patch	-0.77479	-0.55500	-0.59110
Hanwha	Power-active	8 h	Depthwise TCN	-0.62178	-0.63182	-0.64313
Hanwha	Full	12 h	Discrete recurrent	-0.27785	-0.26207	-0.31324
Hanwha	Full	12 h	Inverted-variate	-0.27110	-0.30315	-0.28482
Hanwha	Full	12 h	Joint-patch	-0.20469	-0.24152	-0.21510
Hanwha	Full	12 h	Depthwise TCN	-0.17934	-0.19233	-0.18866
Hanwha	Power-active	12 h	Discrete recurrent	-0.62704	-0.60357	-0.69016
Hanwha	Power-active	12 h	Inverted-variate	-0.60275	-0.65849	-0.64942
Hanwha	Power-active	12 h	Joint-patch	-0.52855	-0.43327	-0.48145
Hanwha	Power-active	12 h	Depthwise TCN	-0.46343	-0.56530	-0.50861
Qcells	Full	1 h	Discrete recurrent	-0.13648	-0.30177	-0.82254
Qcells	Full	1 h	Inverted-variate	-0.01945	-0.08940	-0.05842
Qcells	Full	1 h	Joint-patch	-0.04125	-0.14744	-0.20265
Qcells	Full	1 h	Depthwise TCN	-0.19140	-0.08991	0.01064
Qcells	Power-active	1 h	Discrete recurrent	-0.30502	-0.68714	-1.67725
Qcells	Power-active	1 h	Inverted-variate	-0.01510	-0.10341	-0.09454
Qcells	Power-active	1 h	Joint-patch	-0.30270	-0.42471	-0.34917
Qcells	Power-active	1 h	Depthwise TCN	-0.46033	-0.15339	0.02249
Qcells	Full	4 h	Discrete recurrent	-0.26981	-0.42263	-1.02588
Qcells	Full	4 h	Inverted-variate	-0.28195	-0.33935	-0.29151
Qcells	Full	4 h	Joint-patch	-0.32374	-0.46845	-0.52214
Qcells	Full	4 h	Depthwise TCN	-0.59271	-0.51875	-0.42368
Qcells	Power-active	4 h	Discrete recurrent	-0.60787	-0.97543	-2.18057
Qcells	Power-active	4 h	Inverted-variate	-0.59994	-0.65309	-0.61191
Qcells	Power-active	4 h	Joint-patch	-0.92710	-1.09891	-1.06507
Qcells	Power-active	4 h	Depthwise TCN	-1.33813	-1.10355	-0.93068
Qcells	Full	8 h	Discrete recurrent	-0.41366	-0.49742	-0.76850
Qcells	Full	8 h	Inverted-variate	-0.44038	-0.45492	-0.37560
Qcells	Full	8 h	Joint-patch	-0.45584	-0.54116	-0.55976
Qcells	Full	8 h	Depthwise TCN	-0.59562	-0.65654	-0.54134
Qcells	Power-active	8 h	Discrete recurrent	-0.95209	-1.17593	-1.69645
Qcells	Power-active	8 h	Inverted-variate	-0.96993	-0.98873	-0.91253
Qcells	Power-active	8 h	Joint-patch	-1.15251	-1.26438	-1.17279

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Qcells	Power-active	8 h	Depthwise TCN	-1.38535	-1.48752	-1.19143
Qcells	Full	12 h	Discrete recurrent	-0.35576	-0.27276	-0.26605
Qcells	Full	12 h	Inverted-variate	-0.28983	-0.33920	-0.20222
Qcells	Full	12 h	Joint-patch	-0.25484	-0.28206	-0.29352
Qcells	Full	12 h	Depthwise TCN	-0.25781	-0.22607	-0.25775
Qcells	Power-active	12 h	Discrete recurrent	-0.85934	-0.71506	-0.62348
Qcells	Power-active	12 h	Inverted-variate	-0.63709	-0.71159	-0.54233
Qcells	Power-active	12 h	Joint-patch	-0.69371	-0.70131	-0.62555
Qcells	Power-active	12 h	Depthwise TCN	-0.68415	-0.55354	-0.55718

Table S6: Primary per-seed range_nRMSE values (%).

Alt text: Per-seed Train-range normalized RMSE values permit within-array comparison of error growth while avoiding unsupported AC-capacity normalization. The denominator is fitted only on Train and differs by array.

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	1 h	Discrete recurrent	3.417	3.495	3.348
Sanyo	Full	1 h	Inverted-variate	4.377	4.553	3.750
Sanyo	Full	1 h	Joint-patch	4.841	4.097	3.622
Sanyo	Full	1 h	Depthwise TCN	3.855	2.897	4.199
Sanyo	Power-active	1 h	Discrete recurrent	4.886	5.064	4.806
Sanyo	Power-active	1 h	Inverted-variate	5.248	5.373	5.103
Sanyo	Power-active	1 h	Joint-patch	6.242	5.331	4.569
Sanyo	Power-active	1 h	Depthwise TCN	5.138	3.969	5.291
Sanyo	Full	4 h	Discrete recurrent	10.678	10.631	10.999
Sanyo	Full	4 h	Inverted-variate	11.232	10.285	10.253
Sanyo	Full	4 h	Joint-patch	11.037	9.630	9.800
Sanyo	Full	4 h	Depthwise TCN	11.177	10.645	10.477
Sanyo	Power-active	4 h	Discrete recurrent	15.678	15.624	16.174
Sanyo	Power-active	4 h	Inverted-variate	15.689	13.418	14.165
Sanyo	Power-active	4 h	Joint-patch	15.722	13.225	13.818
Sanyo	Power-active	4 h	Depthwise TCN	15.555	14.876	14.380
Sanyo	Full	8 h	Discrete recurrent	14.995	14.859	15.962
Sanyo	Full	8 h	Inverted-variate	14.152	13.352	13.375
Sanyo	Full	8 h	Joint-patch	15.922	14.025	13.383
Sanyo	Full	8 h	Depthwise TCN	15.907	14.156	14.616
Sanyo	Power-active	8 h	Discrete recurrent	22.262	22.016	23.740
Sanyo	Power-active	8 h	Inverted-variate	19.693	17.987	18.464
Sanyo	Power-active	8 h	Joint-patch	23.108	19.820	18.950
Sanyo	Power-active	8 h	Depthwise TCN	22.543	19.790	20.490
Sanyo	Full	12 h	Discrete recurrent	14.497	13.634	14.850
Sanyo	Full	12 h	Inverted-variate	12.892	12.609	12.908
Sanyo	Full	12 h	Joint-patch	13.971	12.911	12.381
Sanyo	Full	12 h	Depthwise TCN	14.160	13.145	13.144
Sanyo	Power-active	12 h	Discrete recurrent	21.658	20.284	22.140
Sanyo	Power-active	12 h	Inverted-variate	17.664	16.942	17.751
Sanyo	Power-active	12 h	Joint-patch	20.165	18.243	17.515
Sanyo	Power-active	12 h	Depthwise TCN	19.934	18.301	18.352
Hanwha	Full	1 h	Discrete recurrent	3.308	3.600	4.078

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Hanwha	Full	1 h	Inverted-variate	3.316	3.643	3.646
Hanwha	Full	1 h	Joint-patch	4.297	5.050	3.988
Hanwha	Full	1 h	Depthwise TCN	3.079	2.876	3.081
Hanwha	Power-active	1 h	Discrete recurrent	4.703	5.200	5.835
Hanwha	Power-active	1 h	Inverted-variate	4.414	4.900	4.879
Hanwha	Power-active	1 h	Joint-patch	5.377	5.385	4.841
Hanwha	Power-active	1 h	Depthwise TCN	4.005	3.781	4.082
Hanwha	Full	4 h	Discrete recurrent	9.950	9.847	10.944
Hanwha	Full	4 h	Inverted-variate	9.391	10.114	10.061
Hanwha	Full	4 h	Joint-patch	10.611	9.842	9.711
Hanwha	Full	4 h	Depthwise TCN	9.914	10.049	10.251
Hanwha	Power-active	4 h	Discrete recurrent	14.534	14.489	16.077
Hanwha	Power-active	4 h	Inverted-variate	13.443	14.427	14.173
Hanwha	Power-active	4 h	Joint-patch	15.115	13.740	13.642
Hanwha	Power-active	4 h	Depthwise TCN	13.953	14.083	14.418
Hanwha	Full	8 h	Discrete recurrent	13.866	14.746	15.523
Hanwha	Full	8 h	Inverted-variate	14.432	14.674	15.263
Hanwha	Full	8 h	Joint-patch	15.076	13.155	13.410
Hanwha	Full	8 h	Depthwise TCN	13.677	13.702	13.996
Hanwha	Power-active	8 h	Discrete recurrent	20.444	21.860	23.013
Hanwha	Power-active	8 h	Inverted-variate	20.792	20.965	21.624
Hanwha	Power-active	8 h	Joint-patch	21.836	18.687	19.116
Hanwha	Power-active	8 h	Depthwise TCN	19.431	19.275	19.785
Hanwha	Full	12 h	Discrete recurrent	13.050	13.455	13.927
Hanwha	Full	12 h	Inverted-variate	13.482	13.778	14.191
Hanwha	Full	12 h	Joint-patch	13.640	12.208	12.550
Hanwha	Full	12 h	Depthwise TCN	12.448	12.872	12.929
Hanwha	Power-active	12 h	Discrete recurrent	19.335	20.007	20.718
Hanwha	Power-active	12 h	Inverted-variate	19.289	19.523	19.911
Hanwha	Power-active	12 h	Joint-patch	19.576	17.209	17.819
Hanwha	Power-active	12 h	Depthwise TCN	17.390	18.055	18.030
Qcells	Full	1 h	Discrete recurrent	7.332	12.176	26.361
Qcells	Full	1 h	Inverted-variate	4.999	5.787	5.482
Qcells	Full	1 h	Joint-patch	8.778	9.521	8.151
Qcells	Full	1 h	Depthwise TCN	8.713	5.065	6.899
Qcells	Power-active	1 h	Discrete recurrent	10.642	17.677	38.357
Qcells	Power-active	1 h	Inverted-variate	7.086	7.869	7.731
Qcells	Power-active	1 h	Joint-patch	12.069	13.567	11.555
Qcells	Power-active	1 h	Depthwise TCN	12.556	7.199	9.991
Qcells	Full	4 h	Discrete recurrent	12.822	15.657	30.358
Qcells	Full	4 h	Inverted-variate	13.106	13.598	13.425
Qcells	Full	4 h	Joint-patch	15.535	17.104	16.348
Qcells	Full	4 h	Depthwise TCN	19.986	17.352	17.253
Qcells	Power-active	4 h	Discrete recurrent	18.663	22.746	44.287
Qcells	Power-active	4 h	Inverted-variate	18.886	19.508	19.254
Qcells	Power-active	4 h	Joint-patch	22.167	24.744	23.655
Qcells	Power-active	4 h	Depthwise TCN	29.047	25.202	25.077
Qcells	Full	8 h	Discrete recurrent	17.328	19.187	25.830
Qcells	Full	8 h	Inverted-variate	19.050	18.988	18.448

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Qcells	Full	8 h	Joint-patch	19.578	20.425	19.388
Qcells	Full	8 h	Depthwise TCN	22.556	23.806	21.192
Qcells	Power-active	8 h	Discrete recurrent	25.569	28.174	38.055
Qcells	Power-active	8 h	Inverted-variate	27.709	27.529	26.622
Qcells	Power-active	8 h	Joint-patch	28.378	29.788	28.230
Qcells	Power-active	8 h	Depthwise TCN	33.047	34.912	30.971
Qcells	Full	12 h	Discrete recurrent	15.035	15.374	14.734
Qcells	Full	12 h	Inverted-variate	14.019	14.386	13.512
Qcells	Full	12 h	Joint-patch	14.416	14.164	13.665
Qcells	Full	12 h	Depthwise TCN	13.246	13.134	12.819
Qcells	Power-active	12 h	Discrete recurrent	22.649	22.948	21.922
Qcells	Power-active	12 h	Inverted-variate	20.448	20.811	19.449
Qcells	Power-active	12 h	Joint-patch	20.921	20.748	19.986
Qcells	Power-active	12 h	Depthwise TCN	19.415	19.215	18.634

Table S7: Primary per-seed RMSE_skill values (%).

Alt text: Per-seed RMSE skill relative to Last-value Persistence quantifies improvement beyond local continuity; positive values favor the neural implementation. Every value uses the same origins, labels, and masks as its reference.

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	1 h	Discrete recurrent	55.381	54.361	56.280
Sanyo	Full	1 h	Inverted-variate	42.848	40.540	51.032
Sanyo	Full	1 h	Joint-patch	36.784	46.504	52.699
Sanyo	Full	1 h	Depthwise TCN	49.666	62.169	45.164
Sanyo	Power-active	1 h	Discrete recurrent	55.629	54.019	56.358
Sanyo	Power-active	1 h	Inverted-variate	52.348	51.213	53.664
Sanyo	Power-active	1 h	Joint-patch	43.314	51.588	58.512
Sanyo	Power-active	1 h	Depthwise TCN	53.347	63.962	51.956
Sanyo	Full	4 h	Discrete recurrent	55.058	55.255	53.706
Sanyo	Full	4 h	Inverted-variate	52.724	56.711	56.847
Sanyo	Full	4 h	Joint-patch	53.546	59.469	58.751
Sanyo	Full	4 h	Depthwise TCN	52.955	55.195	55.900
Sanyo	Power-active	4 h	Discrete recurrent	53.089	53.249	51.606
Sanyo	Power-active	4 h	Inverted-variate	53.056	59.850	57.616
Sanyo	Power-active	4 h	Joint-patch	52.958	60.429	58.653
Sanyo	Power-active	4 h	Depthwise TCN	53.457	55.489	56.973
Sanyo	Full	8 h	Discrete recurrent	60.553	60.913	58.010
Sanyo	Full	8 h	Inverted-variate	62.772	64.877	64.814
Sanyo	Full	8 h	Joint-patch	58.114	63.104	64.795
Sanyo	Full	8 h	Depthwise TCN	58.154	62.760	61.550
Sanyo	Power-active	8 h	Discrete recurrent	57.233	57.706	54.394
Sanyo	Power-active	8 h	Inverted-variate	62.169	65.447	64.529
Sanyo	Power-active	8 h	Joint-patch	55.608	61.925	63.597
Sanyo	Power-active	8 h	Depthwise TCN	56.693	61.983	60.637
Sanyo	Full	12 h	Discrete recurrent	67.824	69.741	67.040
Sanyo	Full	12 h	Inverted-variate	71.387	72.014	71.350
Sanyo	Full	12 h	Joint-patch	68.991	71.343	72.520

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Sanyo	Full	12 h	Depthwise TCN	68.572	70.826	70.828
Sanyo	Power-active	12 h	Discrete recurrent	64.239	66.507	63.443
Sanyo	Power-active	12 h	Inverted-variate	70.834	72.025	70.689
Sanyo	Power-active	12 h	Joint-patch	66.704	69.876	71.079
Sanyo	Power-active	12 h	Depthwise TCN	67.084	69.781	69.698
Hanwha	Full	1 h	Discrete recurrent	55.574	51.655	45.229
Hanwha	Full	1 h	Inverted-variate	55.474	51.080	51.031
Hanwha	Full	1 h	Joint-patch	42.297	32.186	46.437
Hanwha	Full	1 h	Depthwise TCN	58.645	61.382	58.622
Hanwha	Power-active	1 h	Discrete recurrent	56.394	51.785	45.895
Hanwha	Power-active	1 h	Inverted-variate	59.070	54.567	54.755
Hanwha	Power-active	1 h	Joint-patch	50.142	50.064	55.112
Hanwha	Power-active	1 h	Depthwise TCN	62.865	64.938	62.148
Hanwha	Full	4 h	Discrete recurrent	57.112	57.557	52.829
Hanwha	Full	4 h	Inverted-variate	59.521	56.405	56.633
Hanwha	Full	4 h	Joint-patch	54.262	57.578	58.142
Hanwha	Full	4 h	Depthwise TCN	57.268	56.686	55.815
Hanwha	Power-active	4 h	Discrete recurrent	54.906	55.046	50.119
Hanwha	Power-active	4 h	Inverted-variate	58.292	55.239	56.026
Hanwha	Power-active	4 h	Joint-patch	53.102	57.371	57.674
Hanwha	Power-active	4 h	Depthwise TCN	56.708	56.307	55.267
Hanwha	Full	8 h	Discrete recurrent	62.830	60.473	58.390
Hanwha	Full	8 h	Inverted-variate	61.315	60.664	59.087
Hanwha	Full	8 h	Joint-patch	59.587	64.737	64.055
Hanwha	Full	8 h	Depthwise TCN	63.338	63.270	62.482
Hanwha	Power-active	8 h	Discrete recurrent	58.419	55.539	53.194
Hanwha	Power-active	8 h	Inverted-variate	57.710	57.359	56.019
Hanwha	Power-active	8 h	Joint-patch	55.587	61.991	61.119
Hanwha	Power-active	8 h	Depthwise TCN	60.478	60.797	59.758
Hanwha	Full	12 h	Discrete recurrent	70.299	69.377	68.303
Hanwha	Full	12 h	Inverted-variate	69.316	68.642	67.703
Hanwha	Full	12 h	Joint-patch	68.956	72.214	71.436
Hanwha	Full	12 h	Depthwise TCN	71.670	70.704	70.574
Hanwha	Power-active	12 h	Discrete recurrent	65.561	64.363	63.098
Hanwha	Power-active	12 h	Inverted-variate	65.643	65.227	64.535
Hanwha	Power-active	12 h	Joint-patch	65.132	69.348	68.261
Hanwha	Power-active	12 h	Depthwise TCN	69.026	67.841	67.885
Qcells	Full	1 h	Discrete recurrent	5.980	-56.142	-238.038
Qcells	Full	1 h	Inverted-variate	35.900	25.792	29.709
Qcells	Full	1 h	Joint-patch	-12.558	-22.089	-4.527
Qcells	Full	1 h	Depthwise TCN	-11.728	35.050	11.532
Qcells	Power-active	1 h	Discrete recurrent	5.756	-56.540	-239.674
Qcells	Power-active	1 h	Inverted-variate	37.248	30.313	31.542
Qcells	Power-active	1 h	Joint-patch	-6.876	-20.143	-2.325
Qcells	Power-active	1 h	Depthwise TCN	-11.186	36.248	11.521
Qcells	Full	4 h	Discrete recurrent	40.800	27.713	-40.160
Qcells	Full	4 h	Inverted-variate	39.491	37.217	38.020
Qcells	Full	4 h	Joint-patch	28.278	21.030	24.524
Qcells	Full	4 h	Depthwise TCN	7.725	19.886	20.342

Array	Scope	Window	Model	Seed 42	Seed 43	Seed 44
Qcells	Power-active	4 h	Discrete recurrent	40.872	27.935	-40.314
Qcells	Power-active	4 h	Inverted-variate	40.165	38.193	38.998
Qcells	Power-active	4 h	Joint-patch	29.769	21.605	25.054
Qcells	Power-active	4 h	Depthwise TCN	7.972	20.152	20.550
Qcells	Full	8 h	Discrete recurrent	49.082	43.619	24.100
Qcells	Full	8 h	Inverted-variate	44.023	44.207	45.793
Qcells	Full	8 h	Joint-patch	42.471	39.982	43.031
Qcells	Full	8 h	Depthwise TCN	33.721	30.048	37.729
Qcells	Power-active	8 h	Discrete recurrent	49.248	44.077	24.463
Qcells	Power-active	8 h	Inverted-variate	44.999	45.357	47.157
Qcells	Power-active	8 h	Joint-patch	43.672	40.872	43.966
Qcells	Power-active	8 h	Depthwise TCN	34.404	30.703	38.525
Qcells	Full	12 h	Discrete recurrent	62.459	61.612	63.211
Qcells	Full	12 h	Inverted-variate	64.995	64.080	66.263
Qcells	Full	12 h	Joint-patch	64.005	64.635	65.881
Qcells	Full	12 h	Depthwise TCN	66.925	67.206	67.992
Qcells	Power-active	12 h	Discrete recurrent	62.674	62.182	63.871
Qcells	Power-active	12 h	Inverted-variate	66.300	65.703	67.948
Qcells	Power-active	12 h	Joint-patch	65.521	65.806	67.062
Qcells	Power-active	12 h	Depthwise TCN	68.003	68.333	69.291

Table S8: Primary horizon-specific evaluation support.

Alt text: Forecast-origin and valid-target counts decrease with horizon and are shared by every method within each array-specific comparison. Full and power-active scopes report their target support separately.

Array	Scope	Window	Forecast origins	Valid target points
Sanyo	Full	1 h	6,589	79,068
Sanyo	Power-active	1 h	3,201	36,190
Sanyo	Full	4 h	5,905	283,440
Sanyo	Power-active	4 h	3,389	129,528
Sanyo	Full	8 h	5,024	482,304
Sanyo	Power-active	8 h	3,629	215,712
Sanyo	Full	12 h	4,160	599,040
Sanyo	Power-active	12 h	3,869	259,345
Hanwha	Full	1 h	6,625	79,500
Hanwha	Power-active	1 h	3,225	36,421
Hanwha	Full	4 h	6,049	290,352
Hanwha	Power-active	4 h	3,522	132,649
Hanwha	Full	8 h	5,281	506,976
Hanwha	Power-active	8 h	3,881	228,420
Hanwha	Full	12 h	4,521	651,024
Hanwha	Power-active	12 h	4,225	287,368
Qcells	Full	1 h	6,463	77,556
Qcells	Power-active	1 h	3,173	36,504
Qcells	Full	4 h	5,491	263,568
Qcells	Power-active	4 h	3,071	123,447
Qcells	Full	8 h	4,227	405,792
Qcells	Power-active	8 h	2,927	184,959

Array	Scope	Window	Forecast origins	Valid target points
Qcells	Full	12 h	2,996	431,424
Qcells	Power-active	12 h	2,800	187,950

Table S9: Complete-12 h-origin sensitivity: Train-range nRMSE (%) for every evaluated prefix. All horizons within a row use origins that possess a complete valid 12 h target.

Alt text: Restricting every prefix to complete 12 h origins changes the retained target distribution; the Qcells one-hour comparison reverses while the original horizon-specific analysis is preserved. This analysis is secondary to the horizon-specific evaluation.

Array	Scope	Model	1 h	4 h	8 h	12 h
Sanyo	Full	Last-value	4.910	18.570	35.390	45.060
Sanyo	Full	Discrete recurrent	2.300	9.580	14.580	14.330
Sanyo	Full	Inverted-variate	3.580	9.480	12.610	12.800
Sanyo	Full	Joint-patch	3.130	9.070	13.460	13.090
Sanyo	Full	Depthwise TCN	2.650	9.500	13.580	13.480
Sanyo	Power-active	Last-value	10.920	33.850	53.830	60.560
Sanyo	Power-active	Discrete recurrent	4.880	19.290	24.900	21.360
Sanyo	Power-active	Inverted-variate	5.250	16.550	19.210	17.450
Sanyo	Power-active	Joint-patch	4.920	16.800	21.750	18.640
Sanyo	Power-active	Depthwise TCN	4.050	17.120	21.370	18.860
Hanwha	Full	Last-value	5.910	20.030	35.460	43.940
Hanwha	Full	Discrete recurrent	3.090	9.200	13.810	13.480
Hanwha	Full	Inverted-variate	3.160	9.070	13.960	13.820
Hanwha	Full	Joint-patch	4.190	9.370	13.240	12.800
Hanwha	Full	Depthwise TCN	2.840	9.360	12.990	12.750
Hanwha	Power-active	Last-value	11.090	32.520	49.820	56.140
Hanwha	Power-active	Discrete recurrent	5.620	16.310	22.310	20.020
Hanwha	Power-active	Inverted-variate	5.090	15.310	21.530	19.570
Hanwha	Power-active	Joint-patch	5.390	15.620	20.520	18.200
Hanwha	Power-active	Depthwise TCN	4.530	15.530	19.740	17.820
Qcells	Full	Last-value	0.070	10.370	28.730	40.050
Qcells	Full	Discrete recurrent	1.380	8.490	15.270	15.050
Qcells	Full	Inverted-variate	2.070	8.290	14.110	13.970
Qcells	Full	Joint-patch	2.910	8.490	14.520	14.080
Qcells	Full	Depthwise TCN	1.290	8.180	13.340	13.070
Qcells	Power-active	Last-value	1.790	32.090	55.410	60.680
Qcells	Power-active	Discrete recurrent	2.190	25.520	28.980	22.510
Qcells	Power-active	Inverted-variate	3.150	23.710	25.740	20.240
Qcells	Power-active	Joint-patch	2.220	23.820	26.710	20.550
Qcells	Power-active	Depthwise TCN	2.010	24.070	24.560	19.090

Table S10: Daily-matched RMSE (kW); the Daily-valid point mask is identical for all methods.

Alt text: On identical Daily-valid target points, Daily Persistence has lower RMSE than the post hoc neural envelope in 22 of 24 comparisons; the exceptions are Hanwha 1 h full and power-active.

Array	Scope	Window	Daily	Last	Discrete	Inverted	Joint-patch	Depthwise
Sanyo	Full	1 h	0.1752	0.4637	0.2072	0.2560	0.2535	0.2212
Sanyo	Power-active	1 h	0.2589	0.6671	0.2980	0.3175	0.3260	0.2907
Sanyo	Full	4 h	0.1595	1.4371	0.6527	0.6417	0.6153	0.6524
Sanyo	Power-active	4 h	0.2359	2.0246	0.9587	0.8738	0.8636	0.9049
Sanyo	Full	8 h	0.1291	2.3001	0.9256	0.8258	0.8753	0.9026
Sanyo	Power-active	8 h	0.1929	3.1534	1.3735	1.1337	1.2495	1.2686
Sanyo	Full	12 h	0.1181	2.7281	0.8677	0.7753	0.7926	0.8166
Sanyo	Power-active	12 h	0.1794	3.6688	1.2940	1.0572	1.1293	1.1427
Hanwha	Full	1 h	0.1852	0.4339	0.2136	0.2061	0.2591	0.1756
Hanwha	Power-active	1 h	0.2734	0.6286	0.3058	0.2758	0.3032	0.2306
Hanwha	Full	4 h	0.1932	1.3501	0.5977	0.5747	0.5864	0.5874
Hanwha	Power-active	4 h	0.2856	1.8787	0.8763	0.8169	0.8257	0.8249
Hanwha	Full	8 h	0.2035	2.1714	0.8582	0.8626	0.8096	0.8043
Hanwha	Power-active	8 h	0.3029	2.8658	1.2691	1.2315	1.1588	1.1364
Hanwha	Full	12 h	0.2031	2.5597	0.7860	0.8054	0.7460	0.7431
Hanwha	Power-active	12 h	0.3053	3.2725	1.1669	1.1410	1.0609	1.0390
Qcells	Full	1 h	0.2805	0.4708	0.9232	0.3274	0.5323	0.4161
Qcells	Power-active	1 h	0.4089	0.6817	1.3417	0.4565	0.7484	0.5986
Qcells	Full	4 h	0.2806	1.3077	1.1842	0.8076	0.9859	1.0987
Qcells	Power-active	4 h	0.4099	1.9054	1.7244	1.1600	1.4200	1.5963
Qcells	Full	8 h	0.2553	2.0549	1.2549	1.1368	1.1953	1.3596
Qcells	Power-active	8 h	0.3780	3.0413	1.8472	1.6472	1.7385	1.9907
Qcells	Full	12 h	0.2436	2.4185	0.9087	0.8436	0.8502	0.7889
Qcells	Power-active	12 h	0.3690	3.6631	1.3587	1.2216	1.2407	1.1523

9 External data, time coordinates, and limitations

Yulara is one 106.6 kW combined facility, not three independent subarray sites. Its file spans 2016-04-01 23:45 to 2026-02-19 23:55. The 2017 segment has 105,122 distinct ordered rows: 105,120 regular five-minute timestamps and two off-grid records. The filename's 2016 label is not the experiment year. The off-grid timestamps are 2017-02-22 15:30:02 (75.150801518689 kW) and 2017-08-07 05:20:01 (-0.047513291616942 kW); temperature and GHI are missing in both. Both have existing floor and ceiling grid records. They are excluded without rounding, filling the regular record, or modifying the source.

Yulara retains its provider local coordinate, with status `PROVIDER_LOCAL_TIME_OFFSET_UNCONFIRMED`. An observed value at raw timestamp t is available at $t + 5$ min. This fixed coordinate has no introduced DST jumps. The official download and system pages identify the facility and power variable. Temperature follows the Celsius field name; GHI retains the provider-native numbers. Detailed aggregation, sensor uncertainty, and invalid-code definitions remain partial. This is a nonblocking documentation limitation under the frozen protocol; it does not authorize guessed unit conversions or absolute cross-site GHI comparisons.

NIST comprises 365 one-minute CSVs with consistent headers and 525,595 distinct ordered timestamps from 2017-01-01 00:00 to 2017-12-31 23:59. Five minutes are absent: 2017-10-20 23:59 and 2017-10-21 00:00, 00:01, 00:02, and 00:03, all at UTC$-$05:00. Parsing, bins, grids, splits, and Daily joins retain this timezone-aware fixed EST/LST coordinate. There are no DST missing or repeated hours.

At an end time T, each variable uses exactly five distinct finite minute observations in $[T - 5\text{min}, T)$. Bins have a midnight origin, five-minute anchor, left closure, right label, and availability T. An incomplete variable has a missing bin value, never a partial mean. The official CSV supplies `Pyra1_Wm2_Avg` directly, and no `Pyra1_mV_Avg` column is present. The AC meter field `PwrMtrP_kW_Avg` supplies input power and labels. `InvPAC_kW_Avg` has 4,712 entries equal to -999 and is excluded from both inputs and labels.

Train-only timing diagnostics compared energy increments with average power and inspected power/GHI correspondence. The unshifted power comparison had mean absolute energy discrepancy 0.164 kWh, versus 0.226 kWh at either adjacent-minute alignment. Power/GHI correlation was 0.972 without a five-minute displacement versus approximately 0.929 with either displacement. These are diagnostic summaries, not predictive results or a Test-based rule search. The conservative interval-end availability remains the frozen rule.

No complete NIST 2017 Ground operation log was obtained. This is recorded as a nonblocking documentation limitation, not evidence that no maintenance, snow, outage, or instrument event occurred. Finite negative power and GHI, and high-GHI/nonpositive-power observations, are retained. NIST has 14,418 negative power and 271,221 negative GHI minute records; 751 records have native GHI above 500 with nonpositive power. Yulara's 2017 raw segment has 52,807 negative power and 50,504 negative GHI records, with zero in the analogous high-GHI/nonpositive-power category. These are descriptive counts, not quality filters or cross-site irradiance comparisons.

Table S11: External raw-field quality inventory for the 2017 segment, before availability shifts and aggregation. Nonnum: non-numeric; Inf: either infinity. Minima, medians, 95th percentiles and maxima use finite values.

Site	Variable	Null	Nonnum	Inf	Min	Median	P95	Max
Yulara	AC power (kW)	64	0	0	-0.059	-0.035	88.133	109.903
Yulara	Temperature (°C)	2950	0	0	-2.400	22.433	36.100	43.210
Yulara	GHI (native)	2950	0	0	-12.872	1.458	970.905	1391.457
NIST	AC meter power (kW)	2127	0	0	-0.647	0.000	199.400	258.400
NIST	Temperature (°C)	2127	0	0	-12.820	15.300	29.260	38.850
NIST	GHI (W/m^2)	2147	0	0	-18.771	-5.703	787.316	1412.584

Table S12: Parameter counts for the two input widths. Changed projections follow from the input width; they are not new models or transferred checkpoints.

Implementation	17 channels	7 channels	Difference
Discrete recurrent	99,362	96,562	-2,800
Inverted-variate	194,960	102,800	-92,160
Joint-patch	148,112	140,432	-7,680
Depthwise TCN	683,024	682,384	-640

10 External preprocessing and analysis details

The seven channels, in order, are historical AC power, temperature, GHI, their three original missingness masks, and the Isolation Forest flag. Non-numeric and non-finite inputs are missing. KNN imputation uses five neighbors and is fitted on Train numerical inputs. Isolation Forest uses the imputed Train values (100 trees, contamination 0.01, random state 42). Masks and flag are appended before the seven-channel MinMaxScaler is fitted on Train. The target scaler uses finite unimputed Train power. Validation and Test only transform; labels are never imputed. Finite negative external targets are retained. Alice's frozen nonnegative-target rule is unchanged.

Train is January–August 2017, Validation September, and Test October–December, interpreted in each fixed coordinate. Windows retain the regular grid, stay within one split, contain 72 historical positions, and start their targets at origin+5 min. Training and Validation use full H144 targets and true finite origin power. Test eligibility is evaluated separately at each prefix. The 1% power-active threshold uses the site's Train maximum and future true labels only for evaluation. Daily uses an exact timestamp-minus-24-hour join, with the same elementwise intersection for every model, seed, and deterministic baseline. Common points do not imply common historical information.

The external primary model is Inverted-variate, selected before external Test predictions. All four implementations and every seed remain in the secondary tables. RMSE skills use matched reference errors and no epsilon when a reference is zero. Bias is mean prediction minus observation; negative R^2 is retained and zero target variance leaves it undefined. Sample SD uses three seeds with divisor two. Deterministic baselines have no stochastic seed repetitions. The external Test split was inspected for data quality, and is a held-out performance split rather than untouched external confirmation.

11 External quantitative evidence

The tables below are generated from `DATA_AUDIT_SUMMARY.csv`, `metrics_per_seed.csv`, and `metrics_summary_mean_sd.csv`. The complete per-seed CSV includes RMSE, MAE, bias, R^2, Train-range nRMSE, both matched skills, origin/target counts, and run metadata. The complete summary CSV includes means and sample SDs for all metrics. They are repository materials; the compact PDF tables preserve all model/seed/horizon/scope combinations without printing the entire long-form data inventory.

Table S13: All 96 external support groups, compressed to 24 rows. Each cell gives origins / valid target points. P: primary; D: Daily-matched. Train/Validation prefix counts are descriptive eligibility inventories; fitting uses 12 h only.

Site	Split	H	P full	P power-active	D full	D power-active
Yulara	train	1 h	69,861 / 838,332	35,109 / 388,983	69,656 / 835,770	34,948 / 387,108
Yulara	train	4 h	69,717 / 3,346,416	43,727 / 1,551,419	69,548 / 3,337,032	43,566 / 1,544,738
Yulara	train	8 h	69,525 / 6,674,400	55,170 / 3,091,616	69,404 / 6,657,936	55,009 / 3,080,543

Site	Split	H	P full	P power-active	D full	D power-active
Yulara	train	12 h	69,333 / 9,983,952	66,248 / 4,623,382	69,260 / 9,962,712	66,098 / 4,610,221
Yulara	validation	1 h	8,459 / 101,508	4,329 / 47,952	8,411 / 100,776	4,281 / 47,232
Yulara	validation	4 h	8,315 / 399,120	5,305 / 189,051	8,303 / 396,198	5,293 / 186,177
Yulara	validation	8 h	8,135 / 780,960	6,556 / 371,232	8,135 / 776,430	6,556 / 366,798
Yulara	validation	12 h	8,037 / 1,157,328	7,802 / 547,225	8,037 / 1,152,674	7,802 / 542,715
Yulara	test	1 h	26,412 / 316,944	14,784 / 164,664	26,412 / 316,944	14,784 / 164,664
Yulara	test	4 h	26,376 / 1,266,048	18,063 / 658,095	26,376 / 1,266,048	18,063 / 658,095
Yulara	test	8 h	26,328 / 2,527,488	22,393 / 1,313,250	26,328 / 2,527,488	22,393 / 1,313,250
Yulara	test	12 h	26,280 / 3,784,320	26,192 / 1,963,842	26,280 / 3,784,320	26,192 / 1,963,842
NIST	train	1 h	68,837 / 826,044	34,613 / 380,214	68,615 / 821,094	34,479 / 376,830
NIST	train	4 h	66,958 / 3,213,984	41,646 / 1,457,504	66,789 / 3,196,711	41,510 / 1,445,846
NIST	train	8 h	64,717 / 6,212,832	50,964 / 2,790,328	64,596 / 6,183,604	50,828 / 2,771,505
NIST	train	12 h	62,643 / 9,020,592	59,066 / 4,062,345	62,570 / 8,982,982	58,940 / 4,039,615
NIST	validation	1 h	8,290 / 99,480	4,132 / 45,799	8,290 / 99,240	4,132 / 45,655
NIST	validation	4 h	7,543 / 362,064	4,548 / 159,943	7,543 / 361,151	4,548 / 159,368
NIST	validation	8 h	6,690 / 642,240	5,099 / 274,705	6,690 / 640,575	5,099 / 273,662
NIST	validation	12 h	5,915 / 851,760	5,625 / 364,699	5,915 / 849,433	5,625 / 363,282
NIST	test	1 h	25,219 / 302,628	10,161 / 109,258	25,219 / 300,173	10,158 / 108,291
NIST	test	4 h	24,240 / 1,163,520	12,577 / 413,664	24,240 / 1,154,170	12,574 / 410,266
NIST	test	8 h	23,037 / 2,211,552	15,700 / 775,182	23,037 / 2,194,678	15,697 / 769,372
NIST	test	12 h	21,885 / 3,151,440	18,602 / 1,106,213	21,885 / 3,128,676	18,599 / 1,098,178

Table S14: Primary external mean/sample-SD errors (kW), mean bias (kW), and mean R^2. Complete per-seed values and all metric SDs are in the accompanying CSV.

Site	H	Scope	Model	RMSE	MAE	Bias	R^2
Yulara	1 h	full	Discrete recurrent	12.320 ± 0.369	5.896 ± 0.254	-0.718	0.871
Yulara	1 h	power-active	Discrete recurrent	16.672 ± 1.060	10.086 ± 0.699	-1.920	0.729
Yulara	4 h	full	Discrete recurrent	15.816 ± 0.201	8.184 ± 0.184	-1.600	0.788
Yulara	4 h	power-active	Discrete recurrent	21.271 ± 0.759	13.773 ± 0.686	-4.459	0.560
Yulara	8 h	full	Discrete recurrent	17.866 ± 0.456	9.606 ± 0.284	-2.948	0.729
Yulara	8 h	power-active	Discrete recurrent	24.289 ± 0.859	16.994 ± 0.694	-6.652	0.427
Yulara	12 h	full	Discrete recurrent	17.883 ± 0.759	9.740 ± 0.381	-3.381	0.728
Yulara	12 h	power-active	Discrete recurrent	24.460 ± 1.206	17.601 ± 0.812	-7.243	0.419
Yulara	1 h	full	Inverted-variate	10.289 ± 0.032	5.888 ± 0.282	-0.290	0.910
Yulara	1 h	power-active	Inverted-variate	13.988 ± 0.063	9.455 ± 0.326	-0.920	0.810
Yulara	4 h	full	Inverted-variate	13.368 ± 0.140	8.360 ± 0.147	-0.668	0.848
Yulara	4 h	power-active	Inverted-variate	17.659 ± 0.242	12.870 ± 0.031	-3.286	0.697
Yulara	8 h	full	Inverted-variate	15.699 ± 0.056	10.028 ± 0.067	-1.585	0.791
Yulara	8 h	power-active	Inverted-variate	20.927 ± 0.150	16.078 ± 0.033	-4.723	0.575
Yulara	12 h	full	Inverted-variate	15.805 ± 0.155	10.226 ± 0.084	-2.132	0.788
Yulara	12 h	power-active	Inverted-variate	21.203 ± 0.221	16.519 ± 0.312	-5.435	0.564
Yulara	1 h	full	Joint-patch	12.598 ± 0.259	7.221 ± 0.656	0.574	0.865
Yulara	1 h	power-active	Joint-patch	16.760 ± 0.667	10.831 ± 0.416	-0.461	0.727
Yulara	4 h	full	Joint-patch	14.633 ± 0.085	9.118 ± 0.064	0.129	0.818
Yulara	4 h	power-active	Joint-patch	18.733 ± 0.338	13.176 ± 0.194	-2.683	0.659
Yulara	8 h	full	Joint-patch	16.795 ± 0.338	10.695 ± 0.299	-0.501	0.761
Yulara	8 h	power-active	Joint-patch	21.426 ± 0.282	16.043 ± 0.313	-4.156	0.554
Yulara	12 h	full	Joint-patch	17.012 ± 0.389	10.728 ± 0.381	-1.423	0.754
Yulara	12 h	power-active	Joint-patch	22.151 ± 0.361	16.695 ± 0.477	-5.308	0.524
Yulara	1 h	full	Depthwise TCN	10.562 ± 0.929	5.916 ± 0.902	1.151	0.905
Yulara	1 h	power-active	Depthwise TCN	14.035 ± 0.912	8.941 ± 0.968	1.103	0.808
Yulara	4 h	full	Depthwise TCN	15.383 ± 1.228	10.516 ± 1.150	-0.608	0.798

Site	H	Scope	Model	RMSE	MAE	Bias	R^2
Yulara	4 h	power-active	Depthwise TCN	19.096 ± 1.564	14.277 ± 1.444	-3.816	0.644
Yulara	8 h	full	Depthwise TCN	18.435 ± 1.422	12.851 ± 1.196	-2.877	0.710
Yulara	8 h	power-active	Depthwise TCN	23.493 ± 1.947	18.484 ± 1.664	-7.385	0.462
Yulara	12 h	full	Depthwise TCN	18.875 ± 1.798	13.265 ± 1.285	-4.054	0.696
Yulara	12 h	power-active	Depthwise TCN	24.227 ± 2.457	19.261 ± 1.895	-8.424	0.427
NIST	1 h	full	Discrete recurrent	17.303 ± 0.462	8.521 ± 0.437	-0.530	0.914
NIST	1 h	power-active	Discrete recurrent	28.431 ± 0.837	19.949 ± 1.509	-0.886	0.815
NIST	4 h	full	Discrete recurrent	27.667 ± 0.249	15.150 ± 0.439	1.025	0.776
NIST	4 h	power-active	Discrete recurrent	43.523 ± 0.487	31.982 ± 0.658	-5.539	0.566
NIST	8 h	full	Discrete recurrent	39.847 ± 1.254	22.530 ± 0.713	7.092	0.522
NIST	8 h	power-active	Discrete recurrent	55.504 ± 0.447	42.326 ± 0.203	-0.255	0.286
NIST	12 h	full	Discrete recurrent	44.383 ± 1.235	25.486 ± 0.555	12.007	0.402
NIST	12 h	power-active	Discrete recurrent	61.477 ± 0.774	48.059 ± 0.379	10.867	0.116
NIST	1 h	full	Inverted-variate	17.822 ± 0.113	10.260 ± 0.350	-2.118	0.908
NIST	1 h	power-active	Inverted-variate	28.404 ± 0.093	20.126 ± 0.295	-1.457	0.816
NIST	4 h	full	Inverted-variate	28.504 ± 0.247	17.015 ± 0.635	-0.128	0.762
NIST	4 h	power-active	Inverted-variate	44.434 ± 0.994	32.967 ± 0.723	-5.270	0.548
NIST	8 h	full	Inverted-variate	40.216 ± 0.741	25.093 ± 0.771	7.299	0.513
NIST	8 h	power-active	Inverted-variate	55.675 ± 0.781	43.249 ± 0.454	1.708	0.281
NIST	12 h	full	Inverted-variate	44.617 ± 0.684	28.558 ± 0.456	12.668	0.396
NIST	12 h	power-active	Inverted-variate	60.610 ± 1.033	48.410 ± 0.699	12.595	0.141
NIST	1 h	full	Joint-patch	19.415 ± 0.415	11.401 ± 0.656	-2.959	0.891
NIST	1 h	power-active	Joint-patch	31.167 ± 0.581	23.467 ± 0.262	-6.610	0.778
NIST	4 h	full	Joint-patch	29.529 ± 0.419	18.444 ± 0.354	0.592	0.745
NIST	4 h	power-active	Joint-patch	45.040 ± 1.037	34.363 ± 0.724	-7.829	0.535
NIST	8 h	full	Joint-patch	41.754 ± 0.718	26.312 ± 0.840	9.517	0.475
NIST	8 h	power-active	Joint-patch	55.823 ± 0.518	43.843 ± 0.020	3.042	0.277
NIST	12 h	full	Joint-patch	46.485 ± 0.921	29.300 ± 0.777	14.868	0.344
NIST	12 h	power-active	Joint-patch	62.685 ± 0.781	49.975 ± 0.634	15.056	0.081
NIST	1 h	full	Depthwise TCN	17.736 ± 0.079	9.944 ± 0.311	-0.194	0.909
NIST	1 h	power-active	Depthwise TCN	28.426 ± 0.152	20.542 ± 0.362	-3.481	0.815
NIST	4 h	full	Depthwise TCN	29.116 ± 0.778	19.250 ± 0.882	3.483	0.752
NIST	4 h	power-active	Depthwise TCN	42.774 ± 0.129	32.408 ± 0.163	-4.493	0.581
NIST	8 h	full	Depthwise TCN	41.159 ± 1.806	27.886 ± 1.548	10.772	0.490
NIST	8 h	power-active	Depthwise TCN	54.655 ± 0.514	43.028 ± 0.536	3.178	0.307
NIST	12 h	full	Depthwise TCN	45.286 ± 2.493	30.952 ± 1.858	15.401	0.376
NIST	12 h	power-active	Depthwise TCN	60.163 ± 1.805	48.525 ± 1.404	12.876	0.153

Table S15: Daily-matched external mean/sample-SD errors (kW), mean bias (kW), and mean R^2. Complete per-seed values and all metric SDs are in the accompanying CSV.

Site	H	Scope	Model	RMSE	MAE	Bias	R^2
Yulara	1 h	full	Discrete recurrent	12.320 ± 0.369	5.896 ± 0.254	-0.718	0.871
Yulara	1 h	power-active	Discrete recurrent	16.672 ± 1.060	10.086 ± 0.699	-1.920	0.729
Yulara	4 h	full	Discrete recurrent	15.816 ± 0.201	8.184 ± 0.184	-1.600	0.788
Yulara	4 h	power-active	Discrete recurrent	21.271 ± 0.759	13.773 ± 0.686	-4.459	0.560
Yulara	8 h	full	Discrete recurrent	17.866 ± 0.456	9.606 ± 0.284	-2.948	0.729
Yulara	8 h	power-active	Discrete recurrent	24.289 ± 0.859	16.994 ± 0.694	-6.652	0.427
Yulara	12 h	full	Discrete recurrent	17.883 ± 0.759	9.740 ± 0.381	-3.381	0.728
Yulara	12 h	power-active	Discrete recurrent	24.460 ± 1.206	17.601 ± 0.812	-7.243	0.419
Yulara	1 h	full	Inverted-variate	10.289 ± 0.032	5.888 ± 0.282	-0.290	0.910
Yulara	1 h	power-active	Inverted-variate	13.988 ± 0.063	9.455 ± 0.326	-0.920	0.810
Yulara	4 h	full	Inverted-variate	13.368 ± 0.140	8.360 ± 0.147	-0.668	0.848

Site	H	Scope	Model	RMSE	MAE	Bias	R^2
Yulara	4 h	power-active	Inverted-variate	17.659 ± 0.242	12.870 ± 0.031	-3.286	0.697
Yulara	8 h	full	Inverted-variate	15.699 ± 0.056	10.028 ± 0.067	-1.585	0.791
Yulara	8 h	power-active	Inverted-variate	20.927 ± 0.150	16.078 ± 0.033	-4.723	0.575
Yulara	12 h	full	Inverted-variate	15.805 ± 0.155	10.226 ± 0.084	-2.132	0.788
Yulara	12 h	power-active	Inverted-variate	21.203 ± 0.221	16.519 ± 0.312	-5.435	0.564
Yulara	1 h	full	Joint-patch	12.598 ± 0.259	7.221 ± 0.656	0.574	0.865
Yulara	1 h	power-active	Joint-patch	16.760 ± 0.667	10.831 ± 0.416	-0.461	0.727
Yulara	4 h	full	Joint-patch	14.633 ± 0.085	9.118 ± 0.064	0.129	0.818
Yulara	4 h	power-active	Joint-patch	18.733 ± 0.338	13.176 ± 0.194	-2.683	0.659
Yulara	8 h	full	Joint-patch	16.795 ± 0.338	10.695 ± 0.299	-0.501	0.761
Yulara	8 h	power-active	Joint-patch	21.426 ± 0.282	16.043 ± 0.313	-4.156	0.554
Yulara	12 h	full	Joint-patch	17.012 ± 0.389	10.728 ± 0.381	-1.423	0.754
Yulara	12 h	power-active	Joint-patch	22.151 ± 0.361	16.695 ± 0.477	-5.308	0.524
Yulara	1 h	full	Depthwise TCN	10.562 ± 0.929	5.916 ± 0.902	1.151	0.905
Yulara	1 h	power-active	Depthwise TCN	14.035 ± 0.912	8.941 ± 0.968	1.103	0.808
Yulara	4 h	full	Depthwise TCN	15.383 ± 1.228	10.516 ± 1.150	-0.608	0.798
Yulara	4 h	power-active	Depthwise TCN	19.096 ± 1.564	14.277 ± 1.444	-3.816	0.644
Yulara	8 h	full	Depthwise TCN	18.435 ± 1.422	12.851 ± 1.196	-2.877	0.710
Yulara	8 h	power-active	Depthwise TCN	23.493 ± 1.947	18.484 ± 1.664	-7.385	0.462
Yulara	12 h	full	Depthwise TCN	18.875 ± 1.798	13.265 ± 1.285	-4.054	0.696
Yulara	12 h	power-active	Depthwise TCN	24.227 ± 2.457	19.261 ± 1.895	-8.424	0.427
NIST	1 h	full	Discrete recurrent	17.285 ± 0.460	8.518 ± 0.436	-0.506	0.914
NIST	1 h	power-active	Discrete recurrent	28.412 ± 0.831	19.953 ± 1.511	-0.814	0.815
NIST	4 h	full	Discrete recurrent	27.662 ± 0.242	15.155 ± 0.439	1.079	0.776
NIST	4 h	power-active	Discrete recurrent	43.507 ± 0.504	31.985 ± 0.666	-5.402	0.566
NIST	8 h	full	Discrete recurrent	39.850 ± 1.259	22.536 ± 0.715	7.153	0.522
NIST	8 h	power-active	Discrete recurrent	55.469 ± 0.448	42.304 ± 0.208	-0.115	0.285
NIST	12 h	full	Discrete recurrent	44.388 ± 1.245	25.484 ± 0.560	12.068	0.401
NIST	12 h	power-active	Discrete recurrent	61.445 ± 0.780	48.019 ± 0.384	11.001	0.115
NIST	1 h	full	Inverted-variate	17.785 ± 0.116	10.243 ± 0.354	-2.114	0.909
NIST	1 h	power-active	Inverted-variate	28.364 ± 0.090	20.106 ± 0.289	-1.412	0.816
NIST	4 h	full	Inverted-variate	28.493 ± 0.242	17.008 ± 0.640	-0.092	0.762
NIST	4 h	power-active	Inverted-variate	44.420 ± 0.982	32.960 ± 0.707	-5.152	0.547
NIST	8 h	full	Inverted-variate	40.207 ± 0.751	25.096 ± 0.779	7.373	0.513
NIST	8 h	power-active	Inverted-variate	55.612 ± 0.784	43.213 ± 0.458	1.883	0.281
NIST	12 h	full	Inverted-variate	44.609 ± 0.690	28.550 ± 0.462	12.745	0.395
NIST	12 h	power-active	Inverted-variate	60.544 ± 1.041	48.352 ± 0.704	12.775	0.141
NIST	1 h	full	Joint-patch	19.393 ± 0.415	11.397 ± 0.660	-2.942	0.891
NIST	1 h	power-active	Joint-patch	31.142 ± 0.583	23.461 ± 0.262	-6.554	0.778
NIST	4 h	full	Joint-patch	29.536 ± 0.426	18.453 ± 0.362	0.628	0.745
NIST	4 h	power-active	Joint-patch	45.048 ± 1.027	34.383 ± 0.719	-7.722	0.534
NIST	8 h	full	Joint-patch	41.763 ± 0.726	26.320 ± 0.848	9.578	0.475
NIST	8 h	power-active	Joint-patch	55.792 ± 0.508	43.822 ± 0.018	3.186	0.277
NIST	12 h	full	Joint-patch	46.495 ± 0.924	29.300 ± 0.780	14.934	0.343
NIST	12 h	power-active	Joint-patch	62.653 ± 0.785	49.933 ± 0.638	15.204	0.080
NIST	1 h	full	Depthwise TCN	17.715 ± 0.080	9.941 ± 0.315	-0.189	0.909
NIST	1 h	power-active	Depthwise TCN	28.396 ± 0.153	20.534 ± 0.369	-3.465	0.816
NIST	4 h	full	Depthwise TCN	29.110 ± 0.785	19.262 ± 0.891	3.525	0.752
NIST	4 h	power-active	Depthwise TCN	42.745 ± 0.139	32.398 ± 0.174	-4.396	0.581
NIST	8 h	full	Depthwise TCN	41.155 ± 1.819	27.899 ± 1.559	10.841	0.489
NIST	8 h	power-active	Depthwise TCN	54.599 ± 0.526	42.991 ± 0.543	3.309	0.307
NIST	12 h	full	Depthwise TCN	45.279 ± 2.509	30.953 ± 1.869	15.466	0.375
NIST	12 h	power-active	Depthwise TCN	60.100 ± 1.824	48.465 ± 1.414	13.007	0.153

Table S16: External primary RMSE (kW) for every seed. The repository metric tables additionally provide every per-seed MAE, bias, R^2, nRMSE and matched skill.

Site	Model	H	Scope	Seed 42	Seed 43	Seed 44
Yulara	Discrete recurrent	1 h	full	12.484	11.898	12.579
Yulara	Discrete recurrent	1 h	power-active	17.212	15.450	17.353
Yulara	Discrete recurrent	4 h	full	15.932	15.583	15.932
Yulara	Discrete recurrent	4 h	power-active	21.662	20.396	21.755
Yulara	Discrete recurrent	8 h	full	18.205	17.348	18.044
Yulara	Discrete recurrent	8 h	power-active	24.878	23.304	24.686
Yulara	Discrete recurrent	12 h	full	18.479	17.028	18.140
Yulara	Discrete recurrent	12 h	power-active	25.369	23.092	24.920
Yulara	Inverted-variate	1 h	full	10.321	10.257	10.290
Yulara	Inverted-variate	1 h	power-active	13.998	13.921	14.045
Yulara	Inverted-variate	4 h	full	13.295	13.279	13.529
Yulara	Inverted-variate	4 h	power-active	17.386	17.745	17.846
Yulara	Inverted-variate	8 h	full	15.666	15.668	15.763
Yulara	Inverted-variate	8 h	power-active	20.753	21.018	21.008
Yulara	Inverted-variate	12 h	full	15.983	15.735	15.697
Yulara	Inverted-variate	12 h	power-active	21.429	21.190	20.988
Yulara	Joint-patch	1 h	full	12.868	12.575	12.352
Yulara	Joint-patch	1 h	power-active	17.427	16.759	16.093
Yulara	Joint-patch	4 h	full	14.707	14.540	14.653
Yulara	Joint-patch	4 h	power-active	18.964	18.890	18.345
Yulara	Joint-patch	8 h	full	17.143	16.469	16.774
Yulara	Joint-patch	8 h	power-active	21.752	21.263	21.263
Yulara	Joint-patch	12 h	full	17.356	16.590	17.089
Yulara	Joint-patch	12 h	power-active	22.454	21.752	22.248
Yulara	Depthwise TCN	1 h	full	11.632	10.090	9.963
Yulara	Depthwise TCN	1 h	power-active	15.088	13.474	13.543
Yulara	Depthwise TCN	4 h	full	16.794	14.797	14.557
Yulara	Depthwise TCN	4 h	power-active	20.876	17.941	18.471
Yulara	Depthwise TCN	8 h	full	20.075	17.672	17.558
Yulara	Depthwise TCN	8 h	power-active	25.697	22.002	22.781
Yulara	Depthwise TCN	12 h	full	20.950	17.914	17.761
Yulara	Depthwise TCN	12 h	power-active	27.037	22.480	23.164
NIST	Discrete recurrent	1 h	full	17.828	16.957	17.124
NIST	Discrete recurrent	1 h	power-active	29.355	27.725	28.213
NIST	Discrete recurrent	4 h	full	27.767	27.383	27.851
NIST	Discrete recurrent	4 h	power-active	43.298	44.081	43.189
NIST	Discrete recurrent	8 h	full	40.001	38.524	41.017
NIST	Discrete recurrent	8 h	power-active	55.020	55.899	55.592
NIST	Discrete recurrent	12 h	full	44.185	43.258	45.705
NIST	Discrete recurrent	12 h	power-active	61.184	60.893	62.355
NIST	Inverted-variate	1 h	full	17.910	17.862	17.695
NIST	Inverted-variate	1 h	power-active	28.415	28.492	28.306
NIST	Inverted-variate	4 h	full	28.594	28.695	28.225
NIST	Inverted-variate	4 h	power-active	43.395	45.375	44.534
NIST	Inverted-variate	8 h	full	40.813	39.387	40.447
NIST	Inverted-variate	8 h	power-active	54.949	55.576	56.501
NIST	Inverted-variate	12 h	full	44.558	43.965	45.329
NIST	Inverted-variate	12 h	power-active	59.779	60.284	61.767
NIST	Joint-patch	1 h	full	19.421	19.827	18.997
NIST	Joint-patch	1 h	power-active	30.787	31.836	30.879
NIST	Joint-patch	4 h	full	29.045	29.762	29.780
NIST	Joint-patch	4 h	power-active	44.920	46.132	44.067

Site	Model	H	Scope	Seed 42	Seed 43	Seed 44
NIST	Joint-patch	8 h	full	41.149	41.565	42.548
NIST	Joint-patch	8 h	power-active	56.207	56.028	55.233
NIST	Joint-patch	12 h	full	46.730	45.467	47.259
NIST	Joint-patch	12 h	power-active	63.489	61.930	62.635
NIST	Depthwise TCN	1 h	full	17.765	17.797	17.646
NIST	Depthwise TCN	1 h	power-active	28.343	28.333	28.601
NIST	Depthwise TCN	4 h	full	28.778	30.005	28.563
NIST	Depthwise TCN	4 h	power-active	42.795	42.891	42.637
NIST	Depthwise TCN	8 h	full	41.116	42.985	39.375
NIST	Depthwise TCN	8 h	power-active	54.724	55.132	54.110
NIST	Depthwise TCN	12 h	full	45.901	47.414	42.542
NIST	Depthwise TCN	12 h	power-active	60.960	61.433	58.096

Table S17: Frozen 24-run matrix, parameter counts, selected Validation epoch/MSE and training seconds. These are historical run records; no training is performed for manuscript preparation.

Site	Model	Seed	Parameters	Best epoch	Val. MSE	Seconds
Yulara	Discrete recurrent	42	96562	8	0.042345	1394.8
Yulara	Discrete recurrent	43	96562	13	0.029118	2175.8
Yulara	Discrete recurrent	44	96562	6	0.042264	1152.9
Yulara	Inverted-variate	42	102800	20	0.015972	198.4
Yulara	Inverted-variate	43	102800	11	0.016846	128.2
Yulara	Inverted-variate	44	102800	10	0.017832	120.1
Yulara	Joint-patch	42	140432	7	0.031496	96.3
Yulara	Joint-patch	43	140432	12	0.027561	138.2
Yulara	Joint-patch	44	140432	8	0.031563	103.3
Yulara	Depthwise TCN	42	682384	1	0.046002	47.9
Yulara	Depthwise TCN	43	682384	10	0.028054	126.4
Yulara	Depthwise TCN	44	682384	9	0.021205	117.3
NIST	Discrete recurrent	42	96562	16	0.016202	1850.3
NIST	Discrete recurrent	43	96562	11	0.016805	1404.9
NIST	Discrete recurrent	44	96562	15	0.016085	2360.2
NIST	Inverted-variate	42	102800	19	0.017901	171.1
NIST	Inverted-variate	43	102800	16	0.018544	150.3
NIST	Inverted-variate	44	102800	24	0.017946	179.4
NIST	Joint-patch	42	140432	24	0.017498	179.5
NIST	Joint-patch	43	140432	21	0.017589	178.6
NIST	Joint-patch	44	140432	18	0.017219	164.6
NIST	Depthwise TCN	42	682384	25	0.017715	184.4
NIST	Depthwise TCN	43	682384	25	0.018257	181.2
NIST	Depthwise TCN	44	682384	25	0.018794	181.9

Table S18: External post hoc descriptive envelope against Daily on identical target points. It is not the prespecified primary model or a deployable strategy.

Site	H	Scope	Envelope member	RMSE	Daily	Skill (%)
Yulara	1 h	full	Inverted-variate	10.289	16.147	+36.28
Yulara	1 h	power-active	Inverted-variate	13.988	22.302	+37.28
Yulara	4 h	full	Inverted-variate	13.368	16.154	+17.25
Yulara	4 h	power-active	Inverted-variate	17.659	22.305	+20.83

Site	H	Scope	Envelope member	RMSE	Daily	Skill (%)
Yulara	8 h	full	Inverted-variate	15.699	16.137	+2.71
Yulara	8 h	power-active	Inverted-variate	20.927	22.286	+6.10
Yulara	12 h	full	Inverted-variate	15.805	16.066	+1.62
Yulara	12 h	power-active	Inverted-variate	21.203	22.201	+4.50
NIST Ground	1 h	full	Discrete recurrent	17.285	42.074	+58.92
NIST Ground	1 h	power-active	Inverted-variate	28.364	69.198	+59.01
NIST Ground	4 h	full	Discrete recurrent	27.662	41.955	+34.07
NIST Ground	4 h	power-active	Depthwise TCN	42.745	69.477	+38.48
NIST Ground	8 h	full	Discrete recurrent	39.850	41.831	+4.74
NIST Ground	8 h	power-active	Depthwise TCN	54.599	69.706	+21.67
NIST Ground	12 h	full	Discrete recurrent	44.388	41.899	-5.94
NIST Ground	12 h	power-active	Depthwise TCN	60.100	69.748	+13.83

12 Evidence verification for the external replications

All 24 completed runs have seven-channel checkpoints and finite saved predictions. The historical M3 audit recorded that, before its manuscript editing, the M1-R data/protocol tests passed 43/43, M2 ordinary tests 17/17, and artifact tests 10/10, with zero failures, errors, or skips. The independent verifier recomputed 10,432 numerical comparisons from saved artifacts without importing the training metric functions. All passed. Checkpoints reproduced saved predictions within the registered numerical tolerance, and the original 96 support groups remained unchanged.

The original 36 Alice checkpoints, completion records, and prediction files were also inspected as 36 complete, correctly identified groups. Manuscript preparation performs no training or checkpoint updates. External source data and both sets of model artifacts remain read-only, with file sizes and nanosecond modification times checked before and after. Original Alice raw files were read with author authorization during the later post hoc review; no source observations were changed. Programmatic paper comparisons distinguish Alice's descriptive focal model from the external prespecified model and from the post hoc envelope; no pooled forty-comparison inference is made.

13 Provider documentation

Official sources checked on September 7, 2026 include the NIST data paper (doi:10.6028/jres.122.040), dataset (doi:10.18434/M3S67G), NIST Data Dictionary v1.0, the Data.gov NIST campus PV catalogue, the DKA Yulara download/system pages, glossary, and download terms. The provider URLs are included in the main bibliography and the frozen external protocol. The missing 2017 log and partial Yulara metadata remain explicitly recorded; unavailable documents are not reconstructed from observed model behavior.

14 Independent evidence verification

An independent NumPy/Pandas verifier reads the saved H144 predictions, labels, timestamps, and validity arrays without importing the production metric or mask functions. It re-derives primary and Daily-matched masks, Persistence forecasts, RMSE, MAE, Train-range nRMSE, ranks, and sample counts. The frozen Alice audit records all 4,414 comparisons passing the registered numerical tolerances; the maximum absolute discrepancy was 8.51×10^{-12} and the maximum relative discrepancy was 3.23×10^{-11}. The verifier confirmed primary wins of 12, 9, 2, and 1 for the Inverted-variate, Depthwise, Joint-patch, and Discrete implementations, respectively, with zero for Last-value Persistence. It also confirmed that Daily Persistence was lower than the post hoc best-of-four neural envelope in 22 comparisons; the two envelope wins were Hanwha H12 full-timeline and power-active.

15 Alice inference measurement details

Parameter counts include all learned tensors. Batch-one latency and batch-256 throughput were measured in evaluation and inference modes on an NVIDIA GeForce RTX 3060 Laptop GPU with float32 inputs of shape $[B, 72, 17]$. Warm-up preceded synchronized repetitions. Timing excludes model loading, disk input/output, data loading, and host-to-device transfer. Peak allocation is reported in MiB. These hardware-specific measurements support only within-environment comparisons.

16 Unmatched-sample diagnostic

Deleting unavailable lag points from Daily Persistence alone changes its target support and can reverse a headline comparison even when each metric is calculated correctly. The matched-support analysis applies one point mask to Daily Persistence, Last-value Persistence, every neural seed, and the labels. Unmatched-support values are therefore treated only as a methodological diagnostic and never as model-performance evidence.